

Advances in the diagnosis and management of aortic stenosis

Aakriti Gupta,¹ Shubhadarshini Pawar,¹ Martin Leon,² Joanna Chikwe,¹ Jeroen J Bax,³ Michael A Borger,⁴ Raj Makkar¹



¹Karsh Center for Interventional Cardiology, Smidt Heart Institute, Cedars-Sinai Medical Center, Los Angeles, CA, USA

²Columbia University Medical Center, New York, NY, USA

³Leiden University Medical Center, Leiden, The Netherlands

⁴University Clinic of Cardiac Surgery, Leipzig Heart Center, Leipzig, Germany

Correspondence to: A Gupta
aakriti.gupta@csmc.edu

Cite this as: *BMJ* 2026;393:e078952
<http://dx.doi.org/10.1136/bmj-2023-078952>

Series explanation: State of the Art Reviews are commissioned on the basis of their relevance to academics and specialists in the US and internationally. For this reason they are written predominantly by US authors.

ABSTRACT

Aortic stenosis, a major contributor to global cardiovascular morbidity, is expected to increase in prevalence as populations age, potentially doubling by 2050. The past decade has witnessed a seminal shift in the diagnosis and management of aortic stenosis, driven largely by innovations in transcatheter therapies and advances in diagnostic precision. Modern echocardiographic techniques, including three dimensional imaging, cardiac computed tomography, and AI driven imaging have transformed patient care, enabling earlier disease detection, more accurate risk assessment, and individualized treatment planning. The advent of transcatheter aortic valve replacement (TAVR), validated across all surgical risk categories, has redefined therapeutic paradigms, emphasizing patient centered decision making by using multidisciplinary heart teams. Despite these breakthroughs, challenges remain, particularly regarding optimal management strategies for bicuspid valve disease, asymptomatic severe aortic stenosis, moderate aortic stenosis, and younger patient populations, who present unique considerations related to long term valve durability and lifetime intervention sequencing. Furthermore, challenges surrounding valve thrombosis, stroke risk, and structural valve degeneration persist. Artificial intelligence might be able to address some of these challenges, potentially refining risk stratification, procedural execution, and long term follow-up. This review critically appraises contemporary diagnostic and therapeutic developments in aortic stenosis, explores emerging evidence, summarizes nuances in American and European guidelines, and identifies pivotal knowledge gaps that require future research to optimize patient outcomes.

Introduction

Aortic stenosis, a substantial contributor to global cardiovascular morbidity, affects millions worldwide; its prevalence escalates with age.^{1 2} The management of aortic stenosis has witnessed an unparalleled shift, bolstered by the advent of transcatheter therapies and evolution of surgical techniques.³⁻¹⁵ These advances have expanded treatment options for patients and reduced recovery times, heralding a new era where patient recovery and quality of life are increasingly considered.

The modern era of diagnosis and management of aortic stenosis is characterized by several key developments. Advanced imaging technologies like three dimensional echocardiography, four dimensional computed tomography, and AI enhanced imaging have improved diagnostic precision in identifying and managing aortic stenosis, enhancing both surgical and interventional planning, as well as personalized patient care.¹⁶⁻²⁰

The heart team model, which brings together a multidisciplinary group to tailor treatment to the individual's unique medical profile, further underscores this shift toward personalized medicine. Moreover, clinical guidelines continue to evolve as professional societies incorporate new evidence, sometimes resulting in differences between European and American recommendations that reflect varying interpretations and approaches to patient care.^{21 22} The contemporary narrative in aortic stenosis management also embraces a long term perspective, recognizing that with interventions now being performed in younger patients, the focus must extend beyond immediate outcomes to encompass lifetime management strategies.^{23 24}

This review is a comprehensive synopsis of the modern diagnosis and management of aortic stenosis incorporating the latest diagnostic tools, therapeutic advances, and evolving clinical trial data. We will also explore the challenges posed by

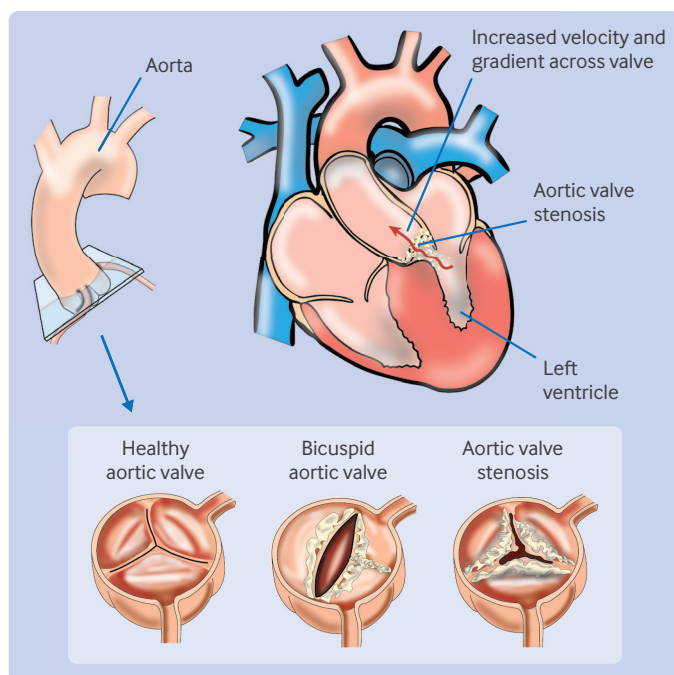


Fig 1 | Illustration of healthy and diseased aortic valves

complex patient populations, nuances of surgical versus transcatheter treatment, considerations for lifetime management of aortic stenosis, and the role of artificial intelligence in shaping the future of aortic stenosis management.

Epidemiology

Aortic stenosis is characterized by progressive narrowing of the aortic valve opening, usually owing to age related calcific degeneration (fig 1). The prevalence of aortic stenosis increases exponentially with age, with 0.2% in those aged 50-59, 1.3% in those aged 60-69, 3.9% in those aged 70-79, and 9.8% in those aged 80-89.²⁵ The prevalence of severe aortic stenosis in individuals aged over 75 is 2% to 5%.²⁶ Importantly, racial differences in aortic stenosis prevalence have been observed, with studies suggesting that African Americans are at substantially lower risk of developing severe aortic stenosis compared with white Americans.^{27 28} The burden of aortic stenosis is expected to grow substantially in the coming decades owing to aging populations. Projections indicate a doubling of severe symptomatic aortic stenosis cases in individuals aged 75 and older by 2050 in both the US and Europe.²⁶

Sources and selection criteria

A comprehensive literature search was conducted using Medline (PubMed), Embase, and the Cochrane Library to identify studies on the contemporary diagnosis and management of aortic stenosis, covering publications from January 1970 to November 2025. The search strategy combined Medical Subject Headings (MeSH) and relevant keywords, including “aortic stenosis,” “transcatheter aortic valve replacement,” “transcatheter aortic

valve implantation,” “echocardiography,” “computed tomography,” “multimodality imaging,” “bioprosthetic,” “artificial intelligence,” and “valve durability.” Therapeutic clinical trials were identified through additional searches of PubMed, Embase, and ClinicalTrials.gov over the same period using the terms “aortic stenosis,” “transcatheter aortic valve replacement,” “surgery,” and “surgical aortic valve replacement.”

Articles were initially screened by reviewing titles and abstracts, followed by assessment of the full text to confirm eligibility. Inclusion criteria encompassed randomized controlled trials, observational studies, meta-analyses, systematic reviews, clinical guidelines, and consensus statements published in peer reviewed journals. Publications in languages other than English were considered if they were accompanied by adequate abstracts or translations. Articles excluded were those published in journals that are not peer reviewed, editorials or commentaries without original data, animal studies, and case reports (see supplementary file).

Quality assessment of studies was based on study design, methodological rigor, and relevance to current clinical practice. Preference was given to studies that influenced current clinical guidelines or contributed novel insights into diagnosis or therapeutic management. Additionally, reference lists of included studies were manually searched to identify further relevant literature. The final selection prioritized high quality, impactful studies that have shaped contemporary understanding and practices in aortic stenosis diagnosis and management.

Pathophysiology

Aortic stenosis is primarily a result of progressive calcification and stiffening of the aortic valve leaflets, driven by an active, osteogenic process.²⁹⁻³² Recent studies have shown that valve interstitial cells play a critical role in initiating this calcification by undergoing osteoblastic differentiation, leading to the formation of calcific nodules. Inflammatory pathways, particularly those involving Lp(a) lipoprotein, oxidized low density lipoprotein, and cytokines, contribute to the pro-calcific environment.^{29-31 33 34} The process of valve calcification is now understood as an active disease, with similarities to atherosclerosis, rather than a passive consequence of aging. This has opened avenues for potential therapeutic interventions targeting Lp(a) lipoprotein and inflammatory pathways, although no pharmacologic treatments have yet proved effective in slowing or reversing the calcific process.³⁵ To date, no medical therapy has convincingly altered the progression of native calcific aortic stenosis. However, even after the valve obstruction is relieved, many patients remain at substantial risk of residual or recurrent heart failure, and adjunctive medical therapy may meaningfully improve outcomes in this phase. In the DapaTAVI trial, dapagliflozin initiated soon after transcatheter aortic valve replacement (TAVR) reduced the 12 month composite of all cause

death or worsening heart failure (15.0% v 20.1%; hazard ratio 0.72, 95% confidence interval 0.55 to 0.95; $P=0.02$), supporting SGLT2 inhibition as a strategy to modify the risk after TAVR rather than a therapy for native aortic stenosis.³⁶

In addition to degenerative aortic stenosis, bicuspid aortic valve disease remains an important cause in younger populations (fig 1).³⁷ Bicuspid aortic valve, a congenital condition affecting 0.5% to 1.0% of the population, predisposes patients to earlier and more aggressive calcification, often necessitating intervention at a younger age.^{37 38} It is also associated with aortic dilation and dissection, requiring close monitoring and, in some cases, combined aortic surgery.^{39–41} Rheumatic aortic stenosis is now uncommon in high income countries owing to the reduced incidence of rheumatic fever, but it remains prevalent in low and middle income countries.⁴² It typically affects both the mitral and aortic valves, resulting from chronic inflammation that leads to leaflet thickening and commissural fusion, distinguishing it from degenerative calcific aortic stenosis.⁴² Additional risk factors for aortic stenosis include mediastinal radiation, renal failure, familial hypercholesterolemia, and calcium metabolism disorders. Genetics also play a role in aortic stenosis, with familial clustering observed in both bicuspid and tricuspid aortic stenosis. Mutations in the NOTCH1 gene have been linked to abnormal calcium deposition in autosomal dominant aortic stenosis, and genome-wide association studies have identified a connection between Lp(a) lipoprotein variants and aortic valve calcification.^{33 43 44}

Advanced imaging in aortic stenosis

The modern era of valvular heart disease diagnosis is characterized by a multimodal, multiparametric approach.⁴⁵ Although echocardiography remains the primary diagnostic tool, advanced techniques and complementary imaging modalities such as cardiac computed tomography and cardiac magnetic resonance play crucial roles in the evaluation of valvular heart disease, enabling more precise quantification, procedural planning, and prognostication.

Transthoracic echocardiography

Transthoracic echocardiography remains the key diagnostic tool for aortic stenosis. Echocardiographic criteria for severe aortic stenosis include an aortic jet velocity ≥ 4.0 m/s, a mean transvalvular gradient ≥ 40 mm Hg, and/or an aortic valve area ≤ 1.0 cm² (indexed aortic valve area ≤ 0.6 cm²/m²).^{22 46} However, in clinical practice, discordant echocardiographic findings—such as low gradients with small aortic valve area—are increasingly recognized, particularly in patients with low flow, low gradient aortic stenosis. This includes both classical (with reduced left ventricular ejection fraction) and paradoxical low flow, low gradient aortic stenosis (with preserved left ventricular ejection fraction but small stroke volume). In such cases, additional imaging and hemodynamic

evaluation are critical. The 2025 European Society of Cardiology/European Association for Cardio-Thoracic Surgery (ESC/EACTS) valvular guidelines recommend a stepwise diagnostic algorithm: confirmation of low stroke volume index (<35 mL/m²), dobutamine stress echocardiography, and quantification of aortic valve calcium by computed tomography when echocardiographic findings remain inconclusive.⁴⁷ Stress echocardiography is useful in asymptomatic patients to unmask symptoms or assess contractile reserve, guiding decisions in borderline cases.^{44 45}

Modern echocardiography goes well beyond basic two dimensional imaging and Doppler. High resolution techniques and adjunct modalities have enhanced diagnostic precision. Three dimensional (3D) echocardiography, particularly transesophageal 3D echocardiography, offers detailed en-face views of valve anatomy that improve the assessment of leaflet pathology and helps guide interventions.⁴⁶ In aortic stenosis, 3D echocardiography permits planimetry of the aortic valve area and better visualization of calcific morphology, complementing Doppler derived gradients.

Myocardial assessment on transthoracic echocardiography also provides complementary information to cardiac magnetic resonance tissue characterization. Measures of left ventricle long axis function—including global longitudinal strain, tissue Doppler S', and mitral annular plane systolic excursion—reflect subendocardial dysfunction and correlate with underlying interstitial fibrosis, often preceding a fall in left ventricular function.^{48 49} Reduced global longitudinal strain measured with strain imaging (speckle tracking echocardiography) in aortic stenosis can therefore identify early left ventricle impairment and has prognostic utility, potentially informing timing of intervention.⁵⁰

Cardiac computed tomography

Cardiac computed tomography has become indispensable in the investigation of aortic stenosis (fig 2). It provides a quantitative calcium score that can confirm the severity of aortic stenosis in cases of low flow, low gradient aortic stenosis where echocardiography measurements are ambiguous.^{51 52} Guidelines note that in low flow, low gradient aortic stenosis with uncertainty, a calcium score >2000 Agatston units in men or >1200 in women strongly indicates true severe aortic stenosis. Current guidelines designate computed tomography angiography as the optimal study for preprocedural planning for any patient being considered for TAVR.^{45 53} The computed tomography scan of the aortic valve complex allows precise annular sizing, which is critical for selecting the appropriate transcatheter valve size and avoiding complications. Unlike two dimensional echocardiography, computed tomography captures the oval shape of the annulus in systole—operators measure annular area and perimeter on computed tomography to decide valve sizing (rather than relying on a single diameter). Annular sizing on computed tomography should be

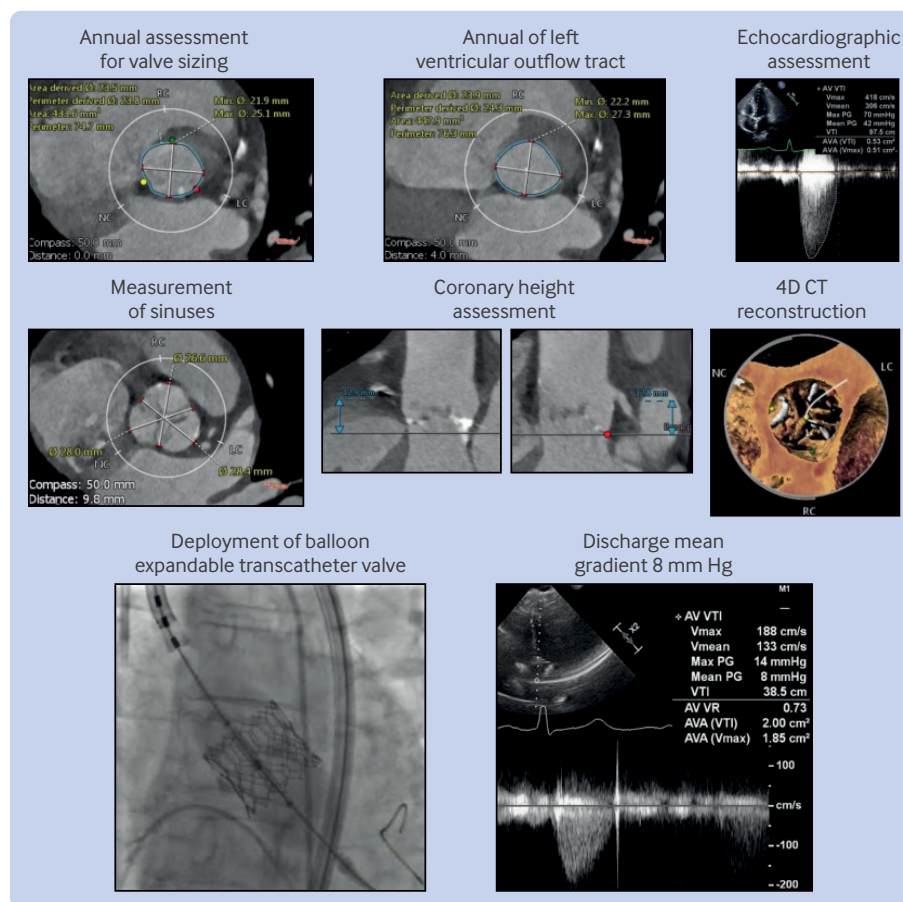


Fig 2 | Preprocedural computed tomography assessment of a 78 year old man for TAVR planning followed by deployment of a balloon expandable valve leading to optimal hemodynamics on echocardiogram at discharge. In this example, the continuous wave Doppler beam is not perfectly aligned with left ventricular outflow/aortic jet. Doppler gradients are angle dependent, so the recorded peak velocity/mean gradient might be underestimated

interpreted in the context of prosthesis design. In contemporary practice, balloon expandable platforms are commonly sized by using approaches based on area (or mean diameter), whereas self-expandable platforms more often emphasize dimensions derived on perimeter to account for non-circular geometry and to guide appropriate oversizing. Notably, sizing based on computed tomography in patients undergoing surgery enables accurate identification of small annulus—a key predictor of postoperative gradients and patient-prosthesis mismatch—and facilitates preoperative planning for annular or root enlargement when indicated. Incorporating sizing derived from computed tomography into surgical aortic valve replacement (SAVR) has been shown to improve prosthesis selection, reduce severe patient-prosthesis mismatch, and minimize interoperator variability, supporting its growing role in contemporary SAVR planning.^{16 54}

Beyond annular dimensions, coronary artery height and sinus of Valsalva width are key computed tomography measurements for TAVR. A low height from the annulus to the coronary ostia (eg, <10–12 mm), especially with bulky leaflets, or narrow sinus diameter raises concern for coronary artery

obstruction when the native leaflets are displaced by the TAVR prosthesis. Computed tomography scans quantify these precisely and even simulate the clearance between a deployed valve frame and the coronaries (eg, virtual “valve to coronary” distances) to predict the risk of coronary obstruction. If the risk is high, operators may plan protective strategies (such as electrosurgical leaflet splitting or coronary protection with stents) or recommend surgery. Similarly, computed tomography identifies bulky calcifications on leaflets or in the left ventricular outflow tract—these are important for risk prediction: extremely calcified, protruding nodules can cause annular rupture or paravalvular leak if not recognized. For example, heavy calcium at the annulus may prompt choosing a self-expanding valve (which exerts lower radial force) or predilatation strategies to minimize the risk of rupture. In patients with bicuspid aortic stenosis, assessment of raphe calcium and annular eccentricity based on computed tomography guides whether TAVR is feasible and what approach to use. Computed tomography of peripheral arteries is used to assess vascular access for TAVR as well. Overall, computed tomography imaging allows creation of an individualized anatomical risk profile (for stroke,

leak, rupture, obstruction, conduction block, etc) before the procedure, enabling the heart team to choose the safest therapy.

Cardiac magnetic resonance imaging

Cardiac magnetic resonance imaging is not routinely used for the assessment of the severity of aortic stenosis because echocardiography and computed tomography generally suffice. However, cardiac magnetic resonance imaging can quantify flow and stroke volume (by using phase contrast imaging), which might be helpful in research settings, particularly for low flow aortic stenosis where standard assessments may be equivocal. It also enables myocardial tissue characterization by late gadolinium enhancement or T1 mapping. Diffuse left ventricular fibrosis owing to prolonged pressure overload from aortic stenosis has prognostic significance; patients with late gadolinium enhancement or high extracellular volume exhibit worse outcomes even after aortic valve replacement.⁵⁵ Indeed, the EVOLVED trial (discussed later) used cardiac magnetic resonance late gadolinium enhancement as an imaging biomarker to guide early intervention in aortic stenosis.⁵⁶

Cardiac damage staging

A newer classification of the severity of aortic stenosis considers the cumulative organ damage from aortic stenosis. This staging system (stages 0-4) is based on the extent of extravalvular cardiac involvement in aortic stenosis.⁵⁷ In stage 0, damage is limited

to the valve (no other structural changes). Stage 1 indicates left ventricular damage, such as concentric hypertrophy, fibrosis, or mild dysfunction. Stage 2 signifies left atrial enlargement or mitral valve involvement, reflecting pressure overload on the left atrium. Stage 3 encompasses pulmonary vasculature and tricuspid valve damage—for instance, pulmonary hypertension and tricuspid regurgitation from prolonged high left-sided pressures. Stage 4 denotes right ventricular failure or dysfunction. This staging has prognostic implications: in a PARTNER 2 trial analysis, one year mortality after aortic valve replacement was only $\approx 4\%$ in patients with stage 0 aortic stenosis, but increased stepwise to 24.5% in patients with stage 4 aortic stenosis.⁵⁷ A recent pooled meta-analysis of observational cohorts and post hoc randomized trial populations, based on reconstructed Kaplan-Meier time to event data, confirmed that higher cardiac damage stage at baseline correlates with worse long term survival after either SAVR or TAVR (eg, 5 year mortality was 57% in stage 4 v 24% in stage 0).⁵⁸ Beyond postintervention outcomes, this staging system also correlates with mortality and major adverse events in conservatively managed patients, emphasizing its broad prognostic relevance.⁵⁹ Clinicians are increasingly using this staging to guide the timing of intervention, aiming to treat earlier in the disease course before patients progress to stage 3-4 damage.

Management

Surgical and transcatheter treatment

The treatment of aortic stenosis has been revolutionized by landmark trials, expanding intervention beyond SAVR to a risk stratified, patient centered approach. TAVR, once limited to patients at high risk, is now a preferred option in broader populations, reshaping the standard of care. SAVR remains essential for younger patients with long life expectancy or complex anatomy, where mechanical or bioprosthetic valves (and less frequently Ross operation or homograft) offer durability. Balloon aortic valvuloplasty, while not a definitive therapy, might serve as a temporary bridge in patients who are hemodynamically unstable. However, it is now infrequently used given that its procedural risk is comparable with TAVR, and it necessitates subsequent valve replacement, thereby exposing patients to cumulative procedural risk.⁶⁰

Surgical aortic valve replacement

SAVR remains the standard of care for many patients with severe aortic stenosis, and the choice between mechanical and bioprosthetic aortic valve replacement represents a fundamental trade-off between durability and the risks associated with lifelong anticoagulation (fig 3). Mechanical valves offer superior longevity, with a reported reoperation rate of 10% at 15 years in the Veterans Affairs randomized trial, compared with 29% for bioprosthetic valves, which deteriorate over time.^{61 62} This pattern is echoed in large administrative and

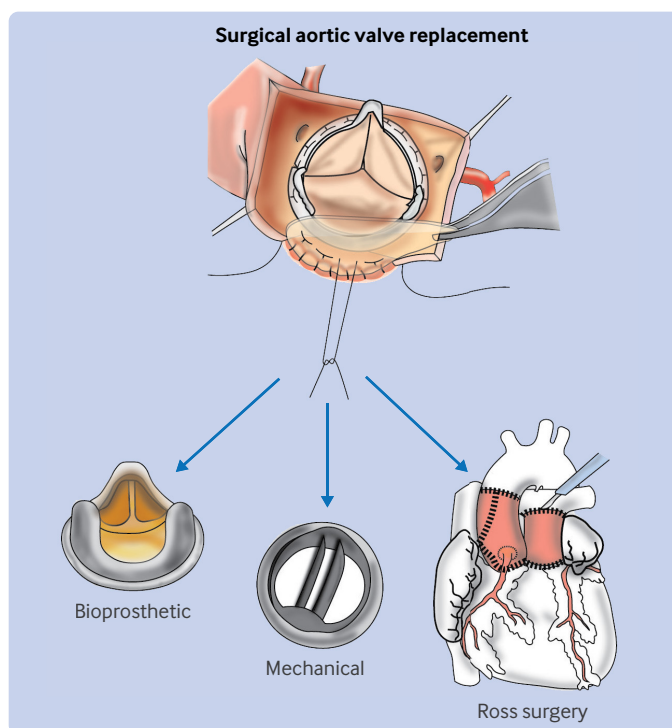


Fig 3 | Illustration of surgical aortic valve replacement by using a mechanical or bioprosthetic valve, Ross surgery (pulmonary autograft), and an example of aortic root enlargement

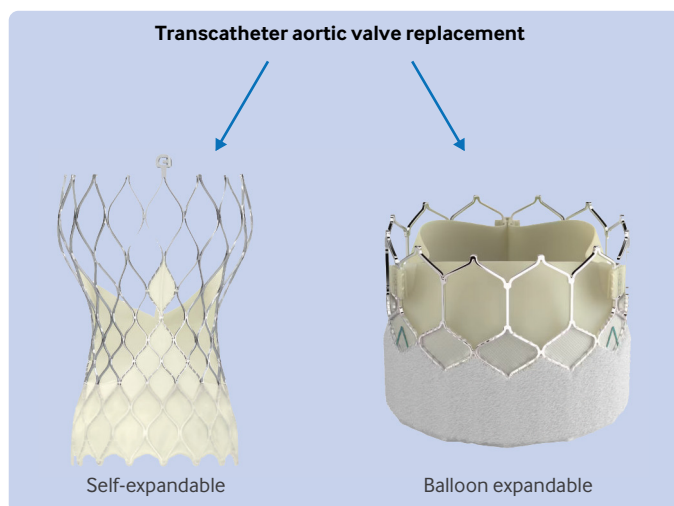


Fig 4 | Illustration of transcatheter aortic valve replacement with bioprostheses that are self-expandable or balloon expandable

clinical datasets, including the Society of Thoracic Surgery (STS) Adult Cardiac Surgery database.^{61 62}

Younger patients (aged <60) typically benefit from mechanical valves, which confer a survival advantage by reducing the need for reintervention. However, this comes at the cost of lifelong warfarin therapy, which increases the risk of major bleeding—reported in 42% of mechanical valve recipients by 11 years versus 26% with bioprostheses.⁶¹⁻⁶⁴ Nevertheless, when anticoagulation is well managed, stroke rates remain comparable between valve types.

In patients aged over 65-70, the survival difference narrows, making bioprosthetic valves more appealing owing to the avoidance of anticoagulation and potential for future valve-in-valve TAVR procedures. For those aged 50-70, the decision must weigh individual factors such as life expectancy, bleeding risk, and procedural planning.

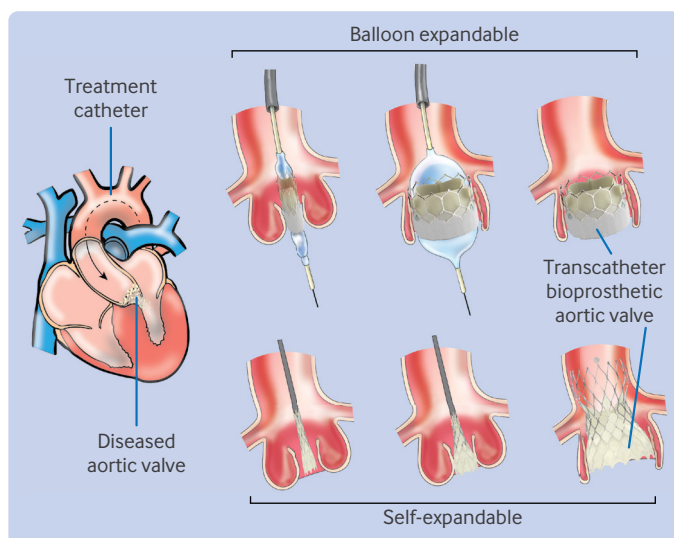


Fig 5 | Illustration of procedural steps for transcatheter aortic valve replacement with bioprostheses that are balloon expandable or self-expandable

A large contemporary STS Adult Cardiac Surgery database analysis linked to the National Death Index (2008-19) evaluated 109 842 patients aged 40-75 undergoing isolated, first time SAVR and found that, after robust risk adjustment, mechanical valves were associated with lower all cause mortality in patients aged ≤60, with no clear survival advantage beyond that age threshold.⁶⁴ The study reported a decline in mechanical valve use over time (from ≈20% to <10% during the study period), underscoring the tension between anticoagulation avoidance and potential long term survival trade-offs in younger patients.

Ross procedure

An alternative for a select group of young patients is the Ross procedure, where the patient's pulmonary valve is relocated to the aortic position, and a homograft replaces the pulmonary valve. This technique provides excellent hemodynamics and avoids anticoagulation, functioning as a "living" aortic valve. However, it is technically complex and involves two valves, requiring surgical expertise. Reinterventions might be needed, particularly for autograft dilatation (in ≈15-20% of adults).⁶⁵ In experienced centers, the Ross procedure has shown excellent long term survival—up to 25 years postoperatively, comparable with the general population.⁶⁶ A meta-analysis of 18 studies (one randomized clinical trial and 17 observational studies) including 3516 patients found a 46% reduction in all cause mortality with the Ross procedure compared with mechanical aortic valve replacement (incidence rate ratio 0.54; 95% confidence interval 0.35 to 0.82; $P=0.004$), with lower stroke and bleeding rates.⁶⁷ These findings have been replicated in national databases from Australia and the UK.^{68 69} Despite this, the Ross procedure remains niche owing to its complexity.

Aortic root enlargement

Other surgical techniques include aortic root enlargement (eg, Nicks or Manouguian procedures) to prevent prosthesis-patient mismatch by enabling the implantation of a larger valve. In cases such as complex endocarditis, a homograft (human donor valve) may be used owing to its infection resistant properties, although availability is limited and durability remains a concern.^{70 71}

Transcatheter aortic valve replacement

TAVR has fundamentally reshaped the management of severe aortic stenosis, shifting from an experimental therapy performed by Cribier in 2002 in an inoperable patient, to a mainstream intervention that rivals surgery across all risk groups (fig 4 and 5). The trajectory of TAVR has been defined by a series of landmark clinical trials that established its safety, efficacy, and durability (fig 6 and 7, table 1).³⁻¹⁵ Over the past two decades, rigorous studies have progressively validated TAVR as a preferred strategy in older patients and those with high surgical risk,

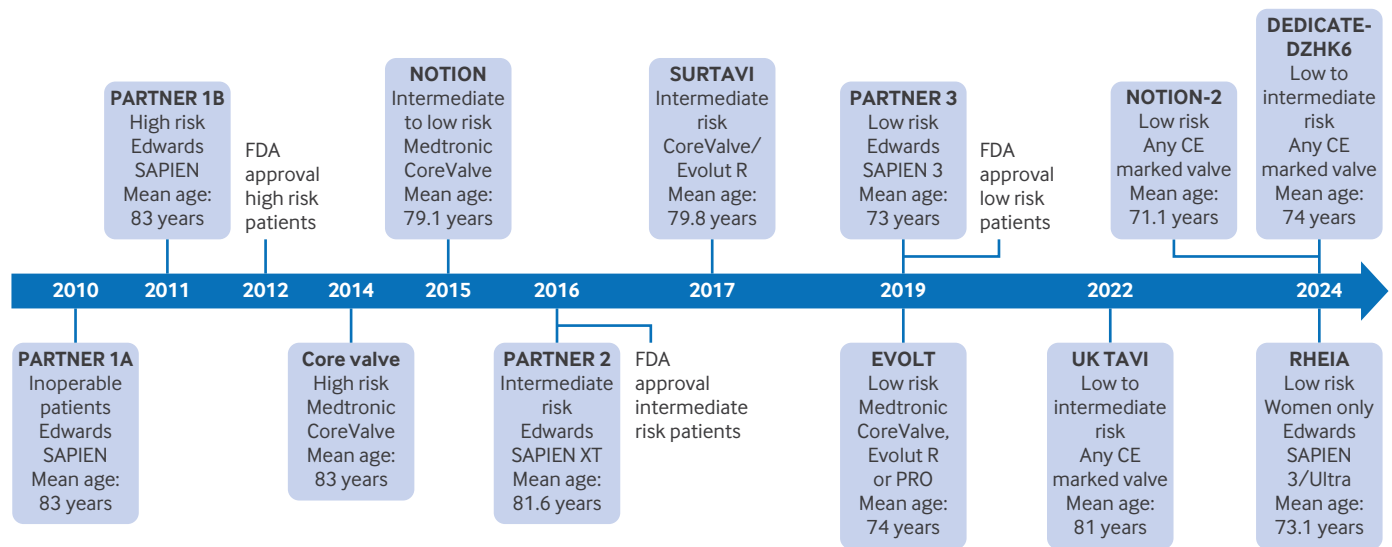


Fig 6 | Timeline of clinical trials of transcatheter aortic valve replacement and surgical aortic valve replacement. FDA=Food and Drug Administration; CE=Conformité Européenne

to a viable option in lower risk populations and even asymptomatic patients.

Bleeding definitions were prespecified and standardized. Several randomized valve trials used VARC-2 criteria, developed by the multidisciplinary Valve Academic Research Consortium (VARC) to harmonize endpoint definitions for randomized controlled trials of valve therapy. Although this supports within-trial comparisons, differences between consensus valve trial definitions and surgical-specific bleeding scales may limit direct comparisons across heterogeneous study designs.

Alongside these developments, the field has confronted challenges such as structural valve degeneration, long term durability, risk of stroke, conduction system abnormalities, and paravalvular leak, shaping the ongoing refinement of this therapy, in addition to expanding to special populations including those with bicuspid morphology or failed bioprosthetic valves.

TAVR prostheses are broadly classified into balloon expandable and self-expandable valves, each with distinct deployment mechanisms, structural properties, and clinical implications (fig 4 and 5). Balloon expandable valves tend to be intra-annular (eg, Sapien series, Edwards Lifesciences; MyVal, Meril Life) and utilize a cobalt chromium or nickel cobalt frame with bovine pericardial leaflets, crimped onto a balloon and expanded forcefully to displace the native calcified leaflets.^{3,72} This provides strong radial force and precise deployment, with an outer skirt that minimizes paravalvular leak. These valves require rapid pacing during deployment for stability, and advancements in delivery systems (now as small as 14 Fr) have improved transfemoral accessibility. For patients with major peripheral arterial disease precluding transfemoral access, alternative access routes such as transaxillary (subclavian), transcarotid, transcaval, and trans-

ortic approaches can be utilized, offering viable pathways with variable safety profiles when performed in experienced centers.⁷³⁻⁷⁴

Self-expandable valves (eg, Evolut FX+ and Evolut PRO+, Medtronic; CoreValve/Evolut R, Medtronic; Portico/Navitor, Abbott; ACURATE neo, Boston Scientific) feature a nitinol frame that gradually expands and conforms to the annulus, often allowing repositionability and retrieval during deployment.^{12,75-77} Most self-expandable valves are supra-annular allowing for improved hemodynamics compared with balloon expandable valves.⁷⁸ The self-expandable design is advantageous in elliptical annuli but can lead to higher paravalvular leak if underexpanded, particularly in bicuspid valves or heavily calcified settings. Although their lower radial force reduces annular rupture risk, it increases conduction disturbances, contributing to higher pacemaker rates compared with balloon expandable valves.⁷⁹⁻⁸¹

Establishing TAVR across the surgical risk spectrum High risk patients

The modern era of TAVR began with the PARTNER 1 trial, which provided the first randomized evidence that transcatheter therapy could offer a lifesaving alternative for patients deemed inoperable for SAVR.³ The trial's cohort B, comparing TAVR to standard medical therapy (including balloon valvuloplasty), showed 20% absolute reduction in mortality at one year (50.7% v 30.7%, $P<0.001$), confirming that TAVR was superior to medical management in patients with no other options.³ Meanwhile, cohort A, which compared TAVR to SAVR in high risk operable patients (STS $\approx 11\%$), found non-inferiority, with similar one year mortality (24.2% TAVR v 26.8% SAVR; $P=0.001$ for non-inferiority, $P=0.62$ for superiority).⁴ Although early stroke risk was a concern, these findings established TAVR as

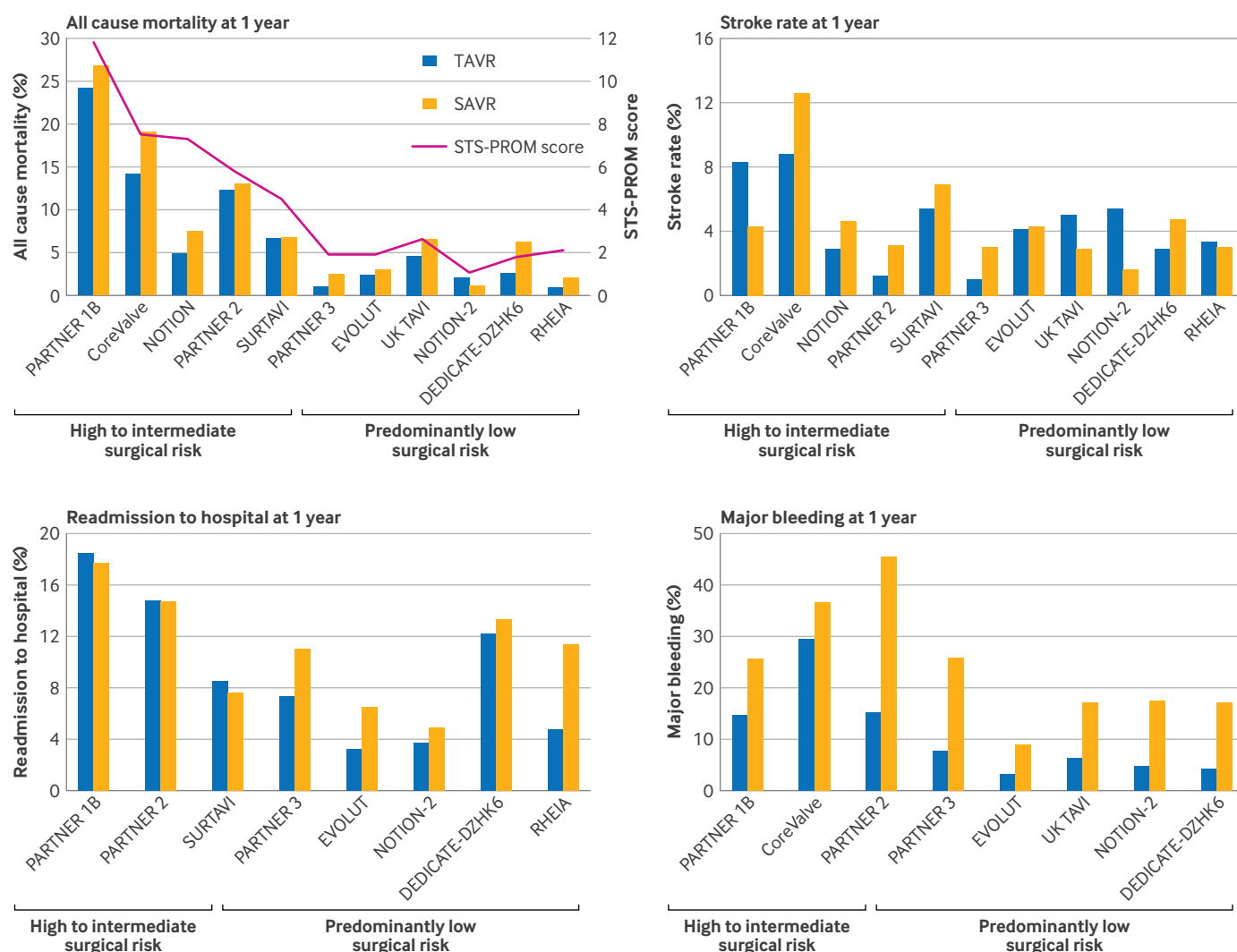


Fig 7 | Outcomes of clinical trials comparing transcatheter aortic valve replacement (TAVR) with surgical aortic valve replacement (SAVR)

a viable therapy, with CE (Conformité Européenne) marking granted in 2007 and subsequent Food and Drug Administration approval for high risk patients in 2012. Concurrently, the CoreValve high risk trial further strengthened TAVR's role in high risk patients, showing superior survival with TAVR compared with SAVR (14.2% v 19.1% 1 year mortality; $P=0.04$ for superiority), with no significant increase in stroke at one year (any stroke 8.8% v 12.6%; $P=0.10$).¹⁵ This suggested that the survival benefit was not offset by neurological complications, bolstering confidence in the broader applicability of TAVR in high risk populations.

Intermediate risk patients

The next logical step was to evaluate TAVR in intermediate risk patients (STS 4% to 8%), who had traditionally undergone SAVR but might benefit from a less invasive approach. This was studied in two major trials: PARTNER 2A and SURTAVI, both of which compared TAVR with SAVR in patients at moderate surgical risk.^{6,8} PARTNER 2A showed non-

inferiority, with similar five year rates of composite of all cause mortality or disabling stroke (47.9% TAVR v 43.4% SAVR).⁷ Similarly, SURTAVI found that TAVR and surgery had similar two year outcomes (death or disabling stroke: 12.6% TAVR v 14.0% SAVR). TAVR's advantages of reduced new onset atrial fibrillation, reduced hospital stays, and faster recovery—were counterbalanced by higher rates of paravalvular leak and pacemaker implantation, but these trade-offs were considered acceptable as device technology improved. With these results, TAVR was approved for intermediate risk patients in 2016, firmly establishing its role beyond the highest risk cohort.

Low risk patients

Sequentially, the performance of TAVR against SAVR in low risk patients (STS <4%) was examined, a group historically managed with surgery owing to concerns about long term durability. The PARTNER 3 trial, evaluating the balloon expandable Sapien 3 valve in low risk patients (mean age 73), delivered

Table 1 | Overview of randomized controlled trials comparing transcatheter aortic valve replacement (TAVR) and surgical aortic valve replacement (SAVR) in patients with aortic stenosis at high, intermediate, and low risk

Study population	Mean age (years)	Risk group	TAVR device*	Primary outcomes	Secondary outcomes	Longest follow-up
PARTNER 1A trial, 2011: placement of aortic transcatheter valve trial						
699 (492 transfemoral (244 TAVR, 248 AVR) v 207 transapical (104 TAVR, 103 AVR))	83	High	Edwards SAPIEN	At 1 year: All cause mortality: 24.2% v 26.8%, P=0.001 for non-inferiority, P=0.62 for superiority Cardiac death: 14% v 13%	At 30 days: Mortality: 3.4% v 6.5%, P=0.07 Vascular complications: 17.0% v 3.8%, P<0.001 Major bleeding: 9.3% v 19.5%, P<0.001 New PPM: 3.8% v 3.6%, P=0.89 All strokes: 5.5% v 2.4%, P=0.04 At 1 year: Vascular complications: 18.0% v 4.8%, P<0.001 Major bleeding: 14.7% v 25.7%, P<0.001 PPM: 5.7% v 5.0%, P=0.68 All strokes: 8.3% v 4.3%, P=0.04 MG: 10.2 mm Hg v 11.5 mm Hg, P=0.008 Repeat hospital stay: 18.2% v 37.7%, P=0.38 Death from any cause or repeat hospital stay: 34.6% v 35.9%, P=0.73	At 5 years: All cause mortality: 67.8% v 62.4%, P=0.76 Repeat hospital stays: 42.3% v 34.2%, P=0.17 All strokes: 15.9% v 14.7%, P=0.35 Moderate or severe AR: 14% vs 1%, P<0.0001 MI: 2.9% v 5.9%, P=0.15 Major vascular complications: 11.9% v 4.7%, P=0.0002 Major bleeding: 26.6% v 34.4%, P=0.003 New PPM: 9.7% v 9.1%, P=0.64
CoreValve trial, 2014: safety and efficacy study of the Medtronic CoreValve system in the treatment of symptomatic severe aortic stenosis in high risk and very high risk subjects who need aortic valve replacement						
747 (390 TAVR, 357 SAVR)	83	High	Medtronic Core Valve	At 1 year: All cause mortality: 14.2% v 19.1%, P<0.001 for non-inferiority, P=0.04 for superiority Stroke: 8.8% v 12.6%, P=0.10 Major stroke: 5.8% v 7%, P=0.59	At 30 days: Vascular complications: 5.9% v 1.7%, P=0.003 New PPM: 19.8% v 7.1%, P<0.001 Bleeding: 13.6% v 35%, P<0.0001 New AF: 11.7% v 30.5%, P<0.0001 Mortality: 3.3% v 4.5% Stroke: 4.9% v 6.2% Major stroke: 3.1% v 3.9%	At 5 years: All cause mortality: 55.3% v 55.4%, P=0.5 Mean days alive and out of the hospital: 1241.4 v 1109.8 days, P=0.006 Stroke: 17.5% v 21.0%, P=0.13 MI: 3.1% v 3.3%, P=0.93 AV reintervention: 3.0% v 1.1%, P=0.04 Endocarditis: 1.8% v 1.7%, P=0.78 New PPM: 38.6% v 22.3%, P<0.001 Moderate SVD: 9.2% v 26.6%, P<0.001 Severe SVD: 0.8% v 1.7%, P=0.32
NOTION trial, 2015: the Nordic aortic valve intervention trial						
280 (145 TAVR, 135 SAVR)	79.1	Intermediate to low	Medtronic Core Valve	Primary composite outcome all cause mortality, MI, stroke At 1 year: Primary: 13.1% v 16.3%, P=0.43 All cause mortality: 4.9% v 7.5%, P=0.38 At 5 years: Primary: 38.0% v 36.3%, P=0.86 All cause mortality: 27.6% v 28.9%, P=0.75	At 30 days: Major vascular complications: 5.6% v 1.5%, P=0.10 Cerebrovascular accident: 2.8% v 3.0%, P=0.94 Cardiogenic shock: 4.2% v 10.4%, P=0.05 Major bleeding: 11.3% v 20.9%, P=0.03 New AF: 16.9% v 57.8%, P<0.0001 New PPM: 34.1% v 1.6%, P<0.001 At 1 year: Moderate to severe AR: 15.7% v 0.9%, P<0.001 NYHA class II: 29.5% v 15.0%, P=0.01 At 5 years: Stroke: 9.0% v 7.4%, P=0.65 PPM: 41.7% v 7.8%, P<0.001 Moderate to severe PVL: 8.2% v 0%, P<0.001 AV reintervention: 2.1% v 0.7%, P=0.35 Endocarditis: 6.2% v 4.4%, P=0.51 MG: 8.2 v 13.7 mm Hg, both P<0.001	At 10 years: Primary: 65.5% v 65.5%, P=0.90 All cause mortality: 62.7% v 64%, P=0.80 Stroke: 9.7% v 16.4%, P=0.1 New PPM: 44.7% v 14.0%, P<0.01 MI: 11% v 8.2%, P=0.4 Moderate to severe SVD: 15.4% v 20.8%, P=0.3 Severe SVD: 1.5% v 10%, P=0.02 AV reintervention: 4.3% v 2.2%, P=0.3 Endocarditis: 7.2% v 7.4%, P=1.0
PARTNER 2, trial, 2016: placement of aortic transcatheter valve trial						
2032 (1011 TAVR, 1021 SAVR)	81.6	Intermediate	Edwards SAPIEN XT	At 2 years: Primary outcome:† 19.3% v 21.1%, P=0.001 All cause mortality: 16.7% v 18.0%, P=0.45 Cardiovascular mortality: 10.1% v 11.3%, P=0.38 Disabling stroke: 6.2% v 6.4%, P=0.83	Within 30 days: Valve embolization: 0.1% v 0% Intensive care unit LOS: 2 v 4 days, P<0.001 Index LOS: 6 v 9 days, P<0.001 All cause mortality: 3.9% v 4.1%, P=0.78 Any neurological event: 6.4% v 6.5%, P=0.94 All strokes: 5.5% v 6.1%, P=0.57 Major vascular complication: 7.9% v 5.0%, P=0.008 Life threatening or disabling bleeding: 10.4% v 43.4%, P<0.0001 New AF: 9.1% v 26.4%, P<0.001 New PPM: 8.5% v 6.9%, P=0.17	At 5 years: Primary outcome:† 47.9% v 43.4%, P=0.21 Freedom from valve intervention: 96.8% v 99.4%, P=0.003 MG: 11.4 v 10.8 mm Hg, P=NS Mild PVL: 17% v 3.5% Moderate to severe PVL: 4.1% v 0.42%, P<0.05 VARC-3 SVD: 9.5% v 3.5%, P<0.001 BVF: 3.7% v 0.8%, P=0.006 All cause BVF: 4.7% v 1.3%, P=0.004

(Continued)

Table 1 | (Continued)

Study population	Mean age (years)	Risk group	TAVR device*	Primary outcomes	Secondary outcomes	Longest follow-up
SURTAVI trial, 2017: safety and efficacy study of the Medtronic CoreValve system in the treatment of severe, symptomatic aortic stenosis in intermediate risk subjects who need aortic valve replacement						
1660	79.8	Intermediate	CoreValve/ Evolut R	At 2 years: Primary outcome:‡ 12.6% v 14.0%, P<0.05 All cause mortality: 11.4% v 11.6% Disabling stroke: 2.6% v 4.5%	At 30 days: All cause mortality: 2.2% v 1.7%, P>0.05 All strokes: 3.4% v 5.6%, P<0.05 Major vascular complication: 6.0% v 1.1%, P<0.05 Life threatening or major bleeding: 12.2% v 9.3%, P>0.05 New PPM: 25.9% v 6.6%, P<0.05 AF: 12.9% v 43.4%, P<0.05 KCCQ change: 18.4% v 5.9%, P<0.05 Mean change in 6-MWT: 35.4 v -14.4, P<0.001	At 5 years: Primary outcome:‡ 31.3% v 30.8%, P=0.85 All cause mortality: 30.0% v 28.7%, P=0.55 AV reintervention: 3.5% v 1.9%, P=0.02 Valve thrombosis: 0.5% v 0.4%, P=0.51 MG: 8.6 v 11.2 mm Hg, P<0.001 EOA: 2.2 v 1.8 cm ² , P<0.001 Moderate to severe PVL: 3.0% v 0.7%, P<0.001
PARTNER 3 trial, 2019: placement of aortic transcatheter valve trial						
950 (496 TAVR, 454 SAVR)	73	Low	Edwards SAPIEN 3	Composite of death, stroke, or repeat hospital stay related to the valve, the procedure, or HF MACE: Within 1 year: 8.5% v 15.1%, P<0.001 2 year: 11.5% v 17.4%, P=0.07	At 30 days: Stroke: 0.6% v 2.4%, P=0.02 New onset AF: 5% v 39.5%, P<0.001 New PPM: 6.5% v 4.0%, P=NS LOS: 3 v 7 days, P<0.001 At 1 year: Mortality or disabling stroke: 1% v 2.9%, P<0.05 Mild paravalvular AR: 29.4% v 2.1%, P<0.05	7 years: MACE: 34.6% v 37.2% All cause mortality: 19.5% v 16.8% Stroke: 8.5% v 8.1% Repeat hospital stay: 20.6% v 23.5% BVF: 6.9% v 7.5%
EVOLUT trial, 2019: Medtronic Evolut transcatheter aortic valve replacement in low risk patients						
1414 (730 TAVR, 684 SAVR)	74	Low	Medtronic CoreValve, Evolut R, or PRO	Primary outcome:‡ At 1 year: 2.9% v 4.6% At 2 years: 5.3% v 6.7%, posterior probability of non-inferiority >0.999	At 30 days: All strokes: 3.4% v 5.6%, P<0.05 Major vascular complication: 3.8% v 3.2%, P>0.05 Life threatening bleeding: 2.4% v 7.5%, P<0.05 New PPM: 17.4% v 6.1%, P<0.05 Moderate to severe PVL: 3.5% v 0.5%, P<0.05 AF: 7.7% v 35.4%, P<0.05 Mean KCCQ change: 20.0 v 9.1, P<0.05 AV reintervention: 1.3% v 1.7%, P=0.63 Clinical or subclinical valve thrombosis: 0.7% v 0.6%, P=0.84 MG: 9.8 v 12.1 mm Hg, P<0.001 AV area: 2.1 v 2.0 cm ² , P<0.001 Mild PVL: 84.7% v 98.4%, P<0.05 At 1 year: New PPM: 19.4% v 6.7%, P<0.001 Valve thrombosis: 0.2% v 0.3%, P=NS Reintervention: 0.7% v 0.6%, P=NS Repeat hospital stay: 3.2% v 6.5%, P=0.01	At 5 years: Primary outcome:‡ 10.7% v 14.1%, P=0.05 All cause mortality: 9% v 12.1%, P=0.07 Disabling stroke: 2.9% v 3.8%, P=0.32 AV reintervention: 1.3% v 1.7%, P=0.63 Clinical or subclinical valve thrombosis: 0.7% v 0.6%, P=0.84 New PPM: 24.6% v 9.9%, P<0.001 MG: 9.8 v 12.1 mm Hg, P<0.001 AV area: 2.1 v 2.0 cm ² , P<0.001 Primary: 15.5% v 16.4%, P=NS All cause mortality: 9.2% v 12.1%, P=NS Disabling stroke: 2.7% v 3.5%, P=NS At 6 years: Primary outcome: ‡ 23.3% v 20.4%, P=0.43 AV reintervention: 5.5% v 3.3%, P=0.07 At 7 years: AV reintervention: 9.8% v 6.0%, P=0.02
UK TAVI trial, 2022: the UK transcatheter aortic valve implantation						
913 (458 TAVR, 455 SAVR)	81	Low to intermediate	Any CE marked valve	At 1 year: All cause mortality: 4.6% v 6.6%, P=0.23	At 1 year: Cardiac death: 2.8% v 3.3%, P=0.69 Stroke: 5.0% v 2.9%, P=0.13 Major bleeding: 6.3% v 17.1%, P<0.001 New PPM: 12.2% v 6.6%, P<0.001	At 1 year: Vascular complications: 4.8% v 1.3%, P<0.001 Moderate AR: 2.3% v 0.6%
NOTION-2 trial, 2024: the Nordic aortic valve intervention trial						
376 (194 TAVR, 176 SAVR)	71.1	Low	Any CE marked valve	At 1 year: Primary outcome:§ 6.4% v 6.6%, P=0.9 All cause mortality: 2.1% v 1.1%, P=0.4	At 1 year: All stroke: 5.4% v 1.6%, P=0.05 Major bleeding: 4.8% v 17.5%, P<0.001 Major vascular complications: 1.6% v 1.6%, P=1.0 New AF: 3.2% v 41.7%, P<0.001 New PPM: 15.1% v 8%, P=0.03 Valve thrombosis: 1.1% v 0%, P=0.2	At 1 year: Moderate or greater PVL: 4.7% v 0%, P=0.005 Severe patient-prosthesis mismatch: 10.1% v 19.4%, P=0.02 AV reintervention: 1.1% v 2.2%, P=0.4 Non-disabling stroke: 3.7 v 0.5%, P=0.03
DEDICATE-DZHK6 trial, 2024: randomized trial of TAVI v SAVR in patients with severe aortic valve stenosis at low to intermediate risk of mortality						
1414 (701 TAVR, 713 SAVR)	74	Low to intermediate	Any CE marked valve	Primary outcome at 1 year:† 5.4% v 10.0%, P<0.001	At 1 year: All cause mortality: 2.6% v 6.2%, P=0.002 Disabling stroke: 1.3% v 3.1%, P=0.02 AF: 12.4% v 30.8%, P<0.001	At 1 year: Major or life threatening bleeding: 4.3% v 17.2%, P<0.001 At least moderate AR: 2.8% v 1.0%, P=0.09

*A variety of SAVR prostheses were used in every listed trial.

†Composite of all cause mortality or stroke.

‡All cause mortality or disabling stroke.

§Composite of all cause death or stroke or repeat hospital stay related to the procedure, valve, or heart failure.

6-MWT=6-minute walk test; AF=atrial fibrillation; AR=aortic regurgitation; AS=aortic stenosis; AV=aortic valve; AVR=aortic valve replacement; BVF=bioprosthetic valve failure; CE=Conformité Européenne (European Conformity); HF=heart failure; EOA=effective orifice area; KCCQ=Kansas City cardiomyopathy questionnaire; LOS=length of stay; MACE=major adverse cardiovascular events; MG=mean gradient; MI=myocardial infarction; NS=not stated; NYHA=New York Heart Association; PPM=permanent pacemaker; PVL=paravalvular leak; SVD=structural valve deterioration; VARC-3=Valve Academic Research Consortium-3

encouraging results: TAVR was not just non-inferior but superior to SAVR at one year, with a composite endpoint (death, stroke, or repeat hospital stay) of 8.5% versus 15.1% ($P=0.001$).¹¹ Importantly, this result was primarily driven by lower repeat hospital stay rates in the TAVR arm. The 30 day stroke rate was lower in the TAVR arm (0.6% v 2.4%), and quality of life improved more rapidly. Most recently, the seven year follow-up results of PARTNER 3 showed no significant differences between TAVR and SAVR for the composite of death, stroke, or repeat hospital stay (34.6% v 37.2%), with comparable mean gradients (13.1 ± 8.5 mm Hg v 12.1 ± 6.3 mm Hg) and similar rates of bioprosthetic valve failure (6.9% v 7.5%).⁸² The EVOLUT low risk trial, which studied the self-expanding CoreValve/Evolut platform in a similar cohort, confirmed non-inferiority, with a two year rate of primary endpoint of death or disabling stroke of 5.3% in TAVR and 6.7% in the SAVR group, establishing non-inferiority of TAVR.¹² At five years, the composite of death or disabling stroke occurred in 15.5% of patients with TAVR versus 16.4% of patients with SAVR, with similar rates of all cause mortality, disabling stroke, and reintervention rates but higher permanent pacemaker implantation after TAVR.⁸³ These findings prompted regulatory approval of TAVR for low risk patients in 2019, fundamentally altering the treatment paradigm for severe aortic stenosis.

We now have five to seven year follow-up results from these trials suggesting ongoing non-inferiority of TAVR compared with SAVR in selected low risk patients.^{13,82,84} At six years, Evolut Low Risk continued to show similar rates of death or disabling stroke after TAVR and SAVR.⁸⁵ However, late follow-up has also raised important durability concerns, including a divergence in valve reintervention, with higher reintervention after TAVR in the available seven year analysis, mainly driven by aortic regurgitation. These findings reinforce that neither TAVR nor SAVR should be viewed as a single uniform therapy; durability might vary according to valve platform, surgical prosthesis, procedural technique, and mode of valve failure. However, longer term follow-up has shown a “catch up” phenomenon, where the initial benefits of TAVR over SAVR diminish over time, suggesting the possibility of worse outcomes with TAVR in longer term follow-up.⁸⁶ This concern becomes especially relevant as TAVR expands into younger patients with longer life expectancies, emphasizing the importance of continued surveillance for long term valve durability.

Specific populations

With TAVR now widely accepted as an attractive alternative across surgical risk categories, recent trials have sought to optimize its use in specific populations. The UK TAVI trial was an investigator initiated, publicly funded, pragmatic trial involving all UK institutions where TAVR was being performed between 2014 and 2018 that enrolled a broad patient population aged over 70 and allowed TAVR

with any commercially available device, making it more representative of the “real world” than previous trials.⁸⁷ It confirmed non-inferiority of TAVR with any commercially available device compared with SAVR for one year mortality (4.6% TAVR v 6.6% SAVR). Meanwhile, the RHEIA trial, the first randomized trial of TAVR versus SAVR exclusively in women (mean age 73), found that TAVR was superior to surgery with lower rates of the combined outcome of mortality, stroke, and repeat hospital stay (8.9% v 15.6%, mostly driven by repeat hospital stays), reflecting unique advantages in women, who often have smaller annuli and higher surgical risk.^{88,89} This result was primarily driven by lower repeat hospital stay rates, with mortality and stroke rates being similar between groups.

Similarly, the DEDICATE-DZHK6 trial, an all comers study in low and intermediate risk patients (mean age 74), reinforced the safety of TAVR across broader patient profiles, showing non-inferiority to SAVR for death or stroke at one year (TAVR 5.4%, SAVR 10.0%, hazard ratio 0.53; 95% confidence interval 0.35 to 0.79), with a numerical trend favoring TAVR for reduced stroke and repeat hospital stay.⁹⁰ However, the primary efficacy outcome will be reported once five year follow-up is complete for this trial. These trials collectively reinforce TAVR's expanding role beyond high risk populations, highlighting its growing applicability across diverse patient groups. However, the average age of included patients in these trials remains over 70 and patients with bicuspid aortic valve disease were excluded in all the above trials.

Head-to-head non-inferiority trials

As the first generation TAVR platforms achieved regulatory approval based on randomized TAVR versus SAVR trials, the evidentiary bar for subsequent devices shifted. Rather than rerunning surgery comparator trials, newer transcatheter systems have typically been evaluated in head-to-head non-inferiority trials against already approved contemporary TAVR valves, aiming to show comparable clinical safety and effectiveness while offering potential advantages in deliverability, hemodynamics, or regurgitation profiles. In the ACURATE IDE randomized trial, the ACURATE neo2 system failed to meet non-inferiority compared with contemporary commercially available control valves, with higher one year rates of the composite of death, stroke, or repeat hospital stay and more prosthetic valve regurgitation.⁹¹ After publication of these findings, this device was withdrawn from commercial use. By contrast, the LANDMARK randomized trial showed that the Myval balloon expandable valve was non-inferior to the contemporary commercial platforms, with similar one year rates of death, stroke, or valve/procedure related repeat hospital stay and comparable hemodynamic performance.^{72,92}

TAVR in bicuspid aortic stenosis

Bicuspid aortic valve is the most common congenital cardiac anomaly, affecting 0.5% to 1%

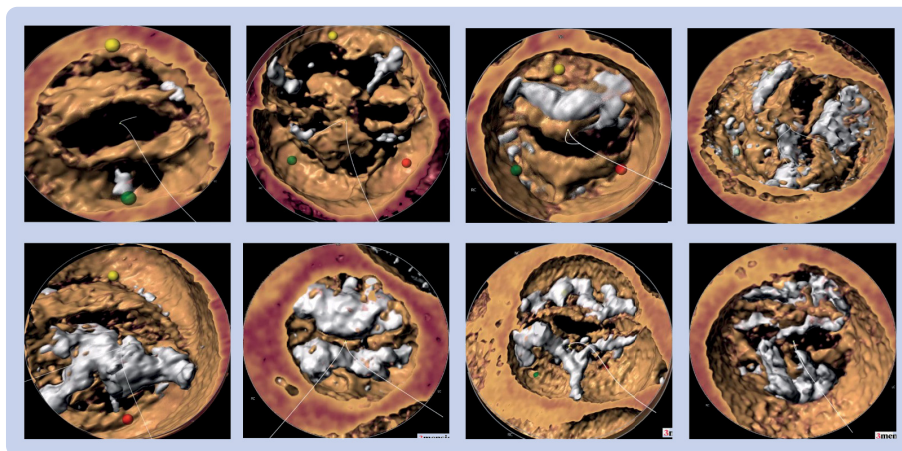


Fig 8 | Spectrum of morphologies of leaflet and commissural calcification in bicuspid aortic stenosis as observed in four dimensional computed tomography rendering

of the population,⁹³ and accounts for a substantial proportion of patients with aortic stenosis undergoing aortic valve replacement, particularly in younger patients.⁹⁴ Despite the regulatory approval and widespread use of TAVR in bicuspid aortic stenosis, these patients were historically excluded from pivotal trials, creating a critical evidence gap. Registry data and propensity matched analyses,^{41 95-100} such as those from the STS/TVT database,⁴¹ have shown comparable short term mortality rates between bicuspid and tricuspid aortic stenosis after TAVR, but higher rates of paravalvular leak and procedural complications, particularly in patients with calcified raphe or extensive leaflet calcification (fig 8). Complex calcification patterns and elliptical annular shape, commonly observed in bicuspid aortic valves, contribute to TAVR complications in this patient group.¹⁰¹ Imaging based analyses have refined risk stratification, identifying high risk bicuspid aortic valve morphologies that predispose to complications such as root injury, severe paravalvular leak, and frame deformation, which in turn may impact long term valve durability and clinical outcomes.^{102 103}

Although studies show that TAVR can be performed successfully in selected patients with bicuspid aortic valve, the lack of long term data beyond two years, particularly in younger individuals with longer life expectancies, remains a major concern. The European NOTION-2 trial, the only randomized controlled trial to include patients with bicuspid aortic valve, suggested a potential signal for harm with TAVR, though it was not imaging guided, was underpowered, and included unfavorable anatomies not typically selected for TAVR in North American practice.¹⁰⁴

Adequately powered randomized clinical trials comparing TAVR with SAVR in bicuspid aortic stenosis are urgently needed to clarify optimal patient selection, long term durability, and outcomes in younger patients. As such, two investigator initiated randomized trials—NAVIGATE in Europe and BELIEVERS in the US—are ongoing to compare

TAVR with SAVR in severe bicuspid aortic stenosis. Their results will clarify comparative effectiveness, durability, and cerebrovascular outcomes in anatomically suitable patients and will help guide future patient selection.¹⁰⁵

TAVR in asymptomatic and moderate aortic stenosis

Several recent randomized trials have reshaped the evidence base for earlier intervention in asymptomatic severe aortic stenosis and have already begun to shift guideline recommendations (table 2). By contrast with prior “watchful waiting” as the default, the 2025 ESC/EACTS guidelines now state that early valve intervention should be considered as an alternative to close surveillance in asymptomatic severe high gradient aortic stenosis when procedural risk is low.⁴⁷ These recommendations are supported by the EARLY TAVR trial, which studied patients (mean age 76) with asymptomatic severe aortic stenosis randomized to early TAVR versus watchful waiting.¹⁰⁶ At a median follow-up of 3.8 years, early TAVR significantly reduced the composite of death, stroke, or unplanned hospital stay (26.8% v 45.3%, hazard ratio 0.5; 95% confidence interval 0.40 to 0.63), mainly by preventing unplanned hospital stays for cardiovascular causes, though mortality was similar. Notably, 87% of the surveillance group eventually required aortic valve replacement, supporting a proactive approach in select low risk asymptomatic patients. In the RECOVERY trial (n=145), which enrolled patients with very severe asymptomatic aortic stenosis, early SAVR reduced the primary composite of operative mortality or cardiovascular death (1% v 15%; hazard ratio 0.09, 95% confidence interval 0.01 to 0.67).¹⁰⁷ Similarly, the AVATAR trial (mean age 67) found that early SAVR reduced major adverse events compared with conservative management, with event rates of 15.2% v 34.7% at 2.7 years (P=0.02) and 23.1% v 46.8% at ≈5 years (P=0.002).^{108 109} The benefit was driven by fewer hospital stays for heart failure and sudden deaths rather than a difference in cardiovascular

Table 2 | Outcomes of randomized controlled trials in moderate and asymptomatic severe aortic stenosis comparing early surgery with conservative care

Study population	Age (mean/ mean $\pm$ SD)	Risk	Device*	Follow-up (years)	Primary outcomes	Secondary outcomes
RECOVERY trial, 2020: early surgery v conventional treatment in very severe aortic stenosis						
145 (73 early surgery, 72 conservative care)	NA (range 20-80)	Asymptomatic severe aortic stenosis	SAVR	6.2	Composite of operative mortality or death from cardiovascular causes: 1% v 15%, P=0.003	All cause death: 7% v 21%, P=0.02 Hospital stay for HF: 0% v 11%, P=0.01 Clinical thromboembolic event: 1% v 6%, P=0.27 Reintervention: 0% v 3%, P=0.33
AVATAR trial, 2021: AVR v conservative treatment in asymptomatic severe aortic stenosis						
157 (78 early SAVR, 79 conservative care)	67	Intermediate to high	SAVR	2.67	Composite of all cause death, acute myocardial infarction, stroke, or unplanned hospitalization for HF: 15.2% v 34.7%, P=0.02	All cause death: 9.54% v 20.11%, P=0.16 Hospital stay for HF: 4.01% v 12.94%, P=0.09 Major bleeding complications: 5.1% v 1.3%, P=0.27 Stroke: 2.6% v 2.5%, P=0.98 Repeated MACEs: 3.8% v 8.9%, P=0.21
EVOLVED trial, 2024: early valve replacement guided by biomarkers of left ventricle decompensation in asymptomatic patients with severe aortic stenosis						
224 (113 early intervention, 111 conservative care)	73 $\pm$ 9	Asymptomatic severe aortic stenosis	SAVR or TAVR	3.5	Composite of all cause death or unplanned aortic stenosis related hospital stay: 18% v 23%, P=0.44	All cause death: 14% v 13%, P=0.59 Unplanned aortic stenosis related hospital stay: 6% v 17%, P=0.02 Stroke: 7% v 13%, P=0.28 Endocarditis: 1% v 3%, P=0.34 Left ventricle systolic impairment: 7% v 10%, P=0.48 NYHA class II-IV at 1 year: 19.7% v 37.9%, P=0.002
TAVR UNLOAD trial, 2024: TAVR to unload the left ventricle in patients with advanced heart failure						
178 (89 TAVR, 89 clinical surveillance)	77	HF with moderate aortic stenosis (patients with reduced ejection fraction)	Sapien3	1.92	Composite of all cause death, disabling stroke, disease related hospital stays, and change from baseline in KCCQ score at 1 year: 48% v 30.9%, Win ratio: 1.55 (95% CI: 1.04 to 2.31), P=0.03 Longest follow-up: 47.6% v 36.3%, Win ratio: 1.31 (95% CI: 0.91 to 1.88), P=0.14	KCCQ overall summary score improvement at 1 year: 12.8 $\pm$ 21.9 v 3.2 $\pm$ 22.8, P=0.018 KCCQ $\geq$ 5-point improvement: 67.2% v 42.4%, P=0.007 Longest follow-up: All cause death: 39.3% v 39.3%, P=0.92 MACE: 51.7% v 57.3%, P=0.37 Stroke: 2.2% v 6.7%, P=0.16 Repeat hospital stay: 38.2% v 44.9%, P=0.35
EARLY TAVR trial, 2025: evaluation of TAVR compared with surveillance for patients with asymptomatic severe aortic stenosis						
901 (455 early TAVR, 446 clinical surveillance)	75.8	Asymptomatic severe aortic stenosis	Sapien3 or Sapien3 Ultra	3.8	Composite of death, stroke, or unplanned hospital stay for cardiovascular causes: 26.8% v 45.3%, P<0.001	All cause death: 8.4% v 9.2%, P=0.72 Stroke: 4.2% v 6.7%, P=0.11 Unplanned hospital stay for cardiovascular causes: 20.9% v 41.7%, P<0.001 Favorable outcome at 2 years (alive with KCCQ score $\geq$ 75): 86.6% v 68.0%, P<0.001 Integrated measures of left ventricle and left atrium health at 2 years: 48.1% v 35.9%, P=0.001 New onset AF: 13.0% v 12.4%, P=0.69 Death or disabling stroke: 9.7% v 11.2%, P=0.50

*A variety of SAVR prostheses were used in every listed SAVR trial.

AF=atrial fibrillation; AVR=aortic valve replacement; CI=confidence interval; HF=heart failure; KCCQ=Kansas City cardiomyopathy questionnaire; MACE=major adverse cardiovascular events; NA=not available; NYHA=New York Heart Association; SD=standard deviation

mortality. However, the EVOLVED trial, which tested early aortic valve replacement (TAVR or SAVR) in patients with asymptomatic severe aortic stenosis with left ventricle fibrosis (detected by cardiac magnetic resonance as evidence of cardiac damage), found no substantial difference in mortality or hospital stays related to aortic stenosis, although this study was not adequately powered.⁵⁶

For patients with moderate aortic stenosis and heart failure with reduced ejection fraction, the TAVR-UNLOAD trial evaluated early TAVR on the background of medical therapy compared with guideline directed medical therapy alone.^{110 111} It did not meet its primary endpoint, as TAVR did not substantially reduce the composite of death and hospital stays for heart failure compared with medical therapy alone. Although there were early functional improvements at one year, by two years the advantage had diminished, possibly owing to crossover to aortic valve replacement. In addition, the trial may not have been adequately powered. The

PROGRESS trial (NCT04889872) and the EXPAND TAVR II trial (NCT05149755) are now investigating the role of early TAVR in moderate aortic stenosis with high risk features such as cardiac damage or dysfunction, which might provide further clarity.

Bioprosthetic valve thrombosis and anticoagulation considerations in TAVR

Bioprosthetic valve thrombosis, a phenomenon historically recognized in surgical aortic valves, has gained attention with the widespread adoption of TAVR and advances in four dimensional computed tomography imaging (fig 9).^{112 113} This condition exists along a spectrum, ranging from subclinical leaflet thrombosis—characterized by hypoattenuated leaflet thickening and reduced leaflet motion seen on computed tomography imaging without manifest symptoms—to clinical valve thrombosis, which manifests as elevated transvalvular gradients, heart failure symptoms, or thromboembolic events.^{114 115} The pathophysiology of leaflet thrombosis in TAVR

is complex, involving altered sinus flow dynamics, valve design factors, features of valve deployment (depth, canting, underexpansion or overexpansion) and patient specific hypercoagulability. Despite initial concerns about its role in the risk of stroke,^{112 113} subsequent observational studies did not confirm these findings.¹¹⁶⁻¹¹⁸ Moreover, randomized clinical trials such as GALILEO and ATLANTIS have shown that routine anticoagulation does not improve long term outcomes and increases major bleeding risk, limiting its use to select high risk cases.^{119 120} Preliminary evidence suggests, however, that hypoattenuated leaflet thickening increases the risk of future structural valve deterioration.¹¹⁷

Current management strategies favor a pragmatic approach, where oral anticoagulation is initiated for confirmed subclinical or symptomatic valve thrombosis,¹¹³ but the choice of agent (a vitamin K antagonist or direct oral anticoagulant) should be left to physician discretion. Routine post-TAVR anticoagulation beyond single or dual antiplatelet therapy remains unjustified in the absence of clinical indications. Ongoing research aims to clarify the long term impact of leaflet thrombosis on valve durability and whether targeted anticoagulation strategies can provide clinical benefit without excess bleeding risk.

Intraprocedural anticoagulation during TAVR is routinely achieved with unfractionated heparin (typically 70 U/kg), with dosing titrated to maintain an activated clotting time of 250-300 seconds. At the end of the procedure, protamine sulfate is administered for heparin reversal at a 1:1 ratio (1 mg protamine per 100 units of heparin), unless contraindicated. This approach is supported by multicenter data demonstrating a reduction in life threatening and major bleeding events, without

an associated increase in stroke or myocardial infarction.^{121 122}

Cerebral protection in TAVR

Despite advancements in TAVR technology and procedural techniques, stroke remains a serious complication, occurring in approximately 2.5% of cases, primarily owing to embolic debris dislodgement during valve deployment.^{123 124} The SENTINEL cerebral embolic protection device is designed to capture embolic material and has been the most widely studied cerebral protection system, and the only one approved by the Food and Drug Administration.

The PROTECTED TAVR trial, the largest randomized study of cerebral protection in TAVR (n=3000), found no significant reduction in overall stroke (2.3% v 2.9%, P=0.30) but demonstrated a 62% relative reduction in disabling stroke (0.5% v 1.3%, P=0.02).^{125 126} Real world data from the STS/ACC TVT Registry, which analyzed over 400 000 TAVR procedures, further suggested a modest reduction in strokes occurring in hospital and disabling stroke with cerebral protection use.¹²⁷ However, after PROTECTED TAVR, utilization of cerebral protection during TAVR has declined, particularly among hospitals that had previously used cerebral embolic protection devices routinely, reflecting ongoing uncertainty about its routine use.^{128 129} This uncertainty was reinforced by the recently published BHF PROTECT-TAVI trial, the largest randomized study to date (n=7635), which was terminated early for futility after interim analysis met prespecified criteria and showed no significant reduction in periprocedural stroke with routine use of cerebral embolic protection (2.1% with cerebral protection

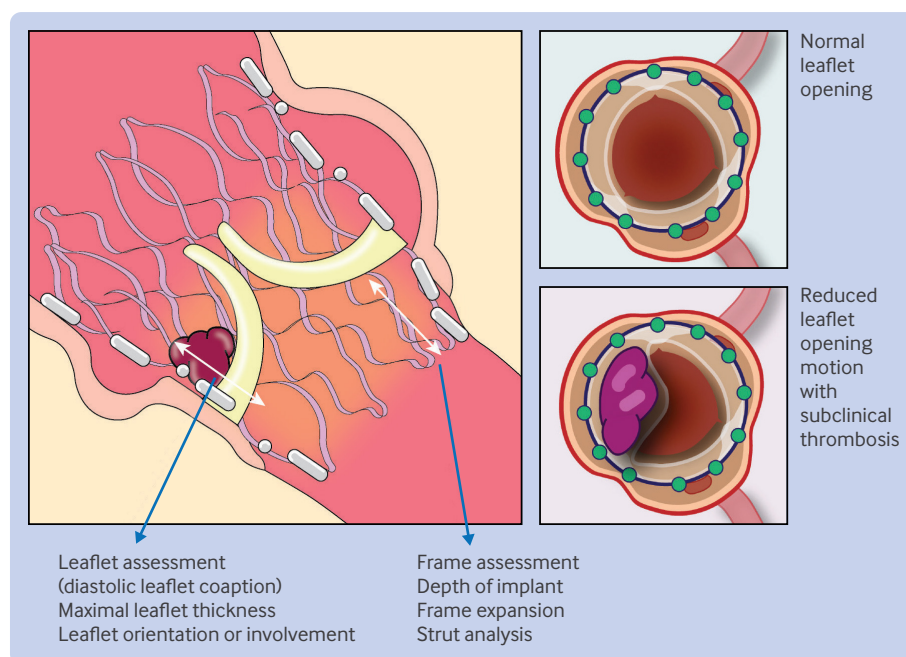


Fig 9 | Illustration of subclinical leaflet thrombosis and reduced leaflet motion

v 2.2% without; risk ratio 0.99, 95% confidence interval 0.73 to 1.34; P=0.94). There was also no significant difference in disabling stroke, mortality, or major vascular complications, and a slightly higher rate of serious adverse events was observed in the cerebral protection group.^{130 131}

Although it might not be warranted in all patients, select high risk groups—such as those with bicuspid valve disease, heavy calcification, previous stroke, or undergoing valve in valve TAVR—might derive greater benefit, and this needs to be the focus of further research.^{41 132}

Severe aortic stenosis and concomitant coronary disease

Coronary artery disease is highly prevalent among patients with severe aortic stenosis, affecting 40% to 70% of cases, and it presents a complex challenge for treatment planning.^{133 134} Current guidelines recommend SAVR with coronary artery bypass grafting as the standard approach, particularly in patients with multivessel or complex coronary artery disease.^{22 135 136} However, the higher perioperative risks and morbidity associated with combined SAVR coronary artery bypass grafting have spurred interest in a less invasive alternative, combining TAVR with percutaneous coronary intervention. The TCW trial, the first and only randomized comparison of physiology guided percutaneous coronary intervention plus TAVR versus SAVR plus coronary artery bypass grafting to date, showed superior one year outcomes with the percutaneous strategy, driven by lower all cause mortality (0% v 10%) and reduced life threatening bleeding (2% v 12%), despite concerns about the need for future revascularization.¹³⁴ Nevertheless, these findings must be interpreted cautiously owing to important methodological limitations, including limited statistical power, short duration of follow-up, low prevalence of complex coronary artery disease, and event rates inconsistent with observational data.

Beyond the choice between surgical and transcatheter approaches, optimal timing and extent of revascularization in TAVR candidates remains uncertain. Given the minimally invasive nature of TAVR and challenges in coronary access postimplantation, coronary artery disease revascularization may sometimes be deferred, especially in patients without angina or ischemic symptoms. Recent results from the multicenter international NOTION-3 trial showed that routine percutaneous coronary intervention—performed primarily before TAVR—substantially reduced the risk of major adverse cardiac events at two years compared with conservative management, although at the cost of increased bleeding.^{137 138} Complementary evidence from a large, real world US registry showed that percutaneous coronary intervention performed within 90 days before TAVR was associated with modestly lower all cause mortality and fewer vascular and bleeding complications compared with

concomitant percutaneous coronary intervention at three years.¹³⁹

Two ongoing trials should further clarify the best strategy. COMPLETE TAVR (NCT04634240) will randomize ≈4000 patients with major coronary artery disease after successful elective transfemoral TAVR to staged percutaneous coronary intervention versus medical therapy alone, with major endpoints including cardiovascular death, myocardial infarction, and heart failure.^{140 141} TAVI PCI (NCT04310046) is comparing percutaneous coronary intervention guided by physiology (fractional flow reserve) versus guided by angiography in this setting. Taken together, the emerging evidence supports a move away from reflex “complete” revascularization for all patients toward percutaneous coronary intervention strategies that are selective, staged, and increasingly guided by physiology and tailored to symptoms, lesion complexity, and anticipated feasibility of future coronary access.

Lifetime management of aortic stenosis

As TAVR expands into younger and lower risk populations, long term durability and sequencing of interventions have become central considerations in the management of aortic stenosis. The choice of index therapy (TAVR or SAVR) is pivotal, because it dictates future treatment options, particularly in patients who will likely require multiple valve interventions over their lifetime.¹³³ Structural valve degeneration remains an unavoidable reality for bioprosthetic valves, occurring due to progressive leaflet calcification, thickening, and mechanical wear and tear,¹⁴² and although the data from trials so far have shown comparable mid-term durability of TAVR to SAVR,^{13 143} data beyond 10-15 years remain limited.

For younger patients, therapy sequencing must be carefully planned to preserve future options. SAVR followed by valve in valve TAVR remains an attractive strategy, because it allows the initial surgical bioprosthesis to serve as a durable anchor for future transcatheter therapies. However, redo TAVR (TAVR in TAVR) is emerging as a viable alternative, with registry data showing acceptable outcomes,¹⁴⁴ though concerns about reduced orifice size, increased transvalvular gradients, and coronary access challenges with each successive implantation remain. These hemodynamic penalties often reflect prosthesis-patient mismatch, particularly in patients with small annuli and after valve in valve procedures. Severe prosthesis-patient mismatch (indexed effective orifice area <0.65 cm²/m²) is less common after native valve TAVR than SAVR (eg, 9% v 28% in PARTNER 2A, but valve in valve implantation increases the risk of severe prosthesis-patient mismatch and has been linked to adverse outcomes in observational data.^{145 146}

In addition, the presence of retained native calcium, compounded by the first transcatheter valve, creates a complex biological and mechanical environment. TAVR explantation, while feasible, is a high risk,

complex procedure, often necessitating aortic root replacement or extensive annular reconstruction, particularly with self-expanding valves that integrate more extensively with native tissue.^{147 148}

Comparative studies suggest that TAVR explantation carries higher morbidity than redo TAVR or redo SAVR.¹⁴⁷⁻¹⁴⁹ Further, if the index valve is a surgical bioprosthesis, studies have shown that valve in valve TAVR is associated with a lower incidence of periprocedural complications and a similar incidence of all cause mortality through two years' follow-up compared with redo SAVR.¹⁵⁰ These data reinforce the importance of initial valve choice and implant technique. Factors such as annular size, coronary height, leaflet calcification, and commissural alignment must be considered at the time of index TAVR to optimize feasibility of future interventions. Future randomized controlled trials will be essential in refining these strategies, providing clearer guidance on optimal therapy sequencing and long term outcomes.

Role of artificial intelligence

AI is increasingly being integrated into the management of aortic stenosis across the care continuum by enhancing diagnostic accuracy, risk stratification, procedural planning, and long term monitoring.²⁰ In addition to improving image interpretation and disease detection, AI can automatically identify cardiac damage phenotypes associated with aortic stenosis, potentially enabling earlier intervention and tailored risk assessment.¹⁵¹ Deep learning approaches have been applied to echocardiography, echocardiography based screening, and computed tomography based quantification (including valve calcification and anatomy relevant to TAVR planning), with the potential to reduce interobserver variability and streamline workflows.^{18 152-164} Risk prediction models based on AI have shown improved performance over traditional scoring systems in forecasting post-TAVR complications, structural valve degeneration, and long term mortality, potentially aiding the selection between TAVR and SAVR in patients at borderline risk.¹⁶⁵ Machine learning algorithms also refine TAVR procedural planning, optimizing annular sizing, commissural alignment, and access routes, while intraprocedural monitoring guided by AI might improve precision and reduce complications such as paravalvular leak and conduction disturbances, and coronary obstruction.^{165 166}

Although most AI applications in aortic stenosis have focused on imaging automation and risk prediction, clinical decisions supported by algorithms have shown outcome impact in pragmatic testing. In a multicenter cluster randomized trial (DETECT-AS), automated provider notifications after echocardiographic identification of severe aortic stenosis increased rates of aortic valve replacement and were associated with improved survival metrics.¹⁶⁷ These findings suggest that the value of data driven tools in aortic stenosis might lie less

in novel model architecture and more in reliably translating diagnostic information into timely referral and treatment.

Emerging treatments

The future of aortic valve disease treatment is being shaped by a wave of investigational therapies that might improve patient outcomes, procedural flexibility, and long term durability. Among these, dedicated transfemoral valve systems for pure aortic regurgitation offer a tailored solution for a population previously underserved by traditional transcatheter devices, which often rely on calcification for anchoring. In the single arm multicenter pivotal investigational device exemption study (ALIGN-AR), a transcatheter device engineered for application in native aortic regurgitation showed high procedural success and robust hemodynamic performance and durability with favorable left ventricular remodeling up to two years.¹⁶⁸⁻¹⁷⁰ These data helped inform the 2025 ESC/EACTS valvular heart disease guidelines, which now include TAVI as a Class IIb, level B treatment option at experienced centers for selected patients with severe symptomatic aortic regurgitation who are ineligible for surgery.⁴⁷ In the US, the Trilogy system has since received Food and Drug Administration approval for symptomatic, severe native tricuspid aortic valve regurgitation in patients with high or greater risk for surgery; in Europe, it carries CE mark approval for both aortic regurgitation and aortic stenosis. Although percutaneous treatment expands access for patients deemed high risk for surgery, it is important to recognize that it complements, but does not replace, surgical aortic valve repair, which remains the optimal treatment.

Other innovations are targeting two key limitations for lifetime management—coronary access after valve in valve TAVR and long term leaflet durability. Strategies such as leaflet modification (including commissural alignment and controlled leaflet splitting) are being developed to preserve coronary access and reduce a major barrier to repeat interventions.¹⁷¹ In parallel, next generation valve designs—including polymeric leaflet platforms and other biomimetic materials intended to resist calcification and potentially reduce reliance on long term anticoagulation—are in early phase clinical evaluation.^{172 173} The fifth generation SAPIEN 3 Ultra Resilia valve incorporates novel anticalcification tissue technology designed to reduce structural valve deterioration. In a large, real world analysis from the Transcatheter Valve Therapy Registry, it demonstrated lower mean gradients, larger effective orifice area, and lower rates of paravalvular leak compared with earlier generation platforms, with similar short term safety outcomes. However, longer term durability data are still awaited as TAVR expands to younger and lower risk patients.¹⁷⁴

New generation cerebral embolic protection devices are also being investigated in randomized

Table 3 | Differences between the American College of Cardiology/American Heart Association (ACC/AHA) 2020 and the European Society of Cardiology/European Association for Cardio-Thoracic Surgery (ESC/EACTS) 2025 guidelines for the management of severe aortic stenosis

Clinical aspect	ACC/AHA	ESC/EACTS
Recommendations based on age	SAVR <65 years or life expectancy >20 years, TAVR ≥80 years or life expectancy <10 years. Shared decision making for ages 65-80	SAVR <75 years (low surgical risk), TAVR ≥70 years or intermediate/high surgical risk. Greater emphasis on lifetime management and coronary access preservation
Selection of tissue v mechanical SAVR	Mechanical valve <50 years, bioprosthesis >65 years. Shared decision making for ages 50-65	Mechanical valve <60 years, bioprosthesis >65 years
Management of bicuspid aortic stenosis	SAVR preferred, TAVR not formally recommended for low risk cases	Class III: TAVR not recommended in low risk/younger patients. Class IIb: TAVR may be considered in older/high risk patients with favorable anatomy
Timing of intervention for symptomatic severe aortic stenosis	Class I AVR for $V_{max} \geq 4$ m/s or mean gradient ≥40 mm Hg. Very severe aortic stenosis defined as $V_{max} \geq 5$ m/s or mean gradient ≥60 mmHg. LFLG aortic stenosis: DSE to confirm severity	Similar class I for severe aortic stenosis. Very severe aortic stenosis also defined as mean gradient ≥60 mm Hg. LFLG aortic stenosis: DSE to confirm severity; CT calcium scoring (sex specific) class I if no contractile reserve
Low flow, low gradient (LFLG) aortic stenosis	Classic LFLG aortic stenosis: AVR class I if DSE confirms severity. Class I even if no contractile reserve if CT calcium score confirms aortic stenosis	Stronger emphasis: CT calcium scoring with sex-specific thresholds (>2000 AU men, >1200 AU women) Class I for confirming severe aortic stenosis if echocardiography/DSE inconclusive.
Paradoxical LFLG aortic stenosis (normal ejection fraction, low stroke volume, low gradient)	AVR if aortic stenosis is most likely cause of symptoms	AVR class I if symptomatic, based on integrated assessment including CT calcium scoring. Noted as debated entity with overlap with HFpEF/moderate aortic stenosis
Asymptomatic severe aortic stenosis	Class I AVR if LVEF <50%. Class IIb AVR if LVEF <60% with progressive decline. BP drop ≥10 mm Hg triggers AVR consideration	Class I AVR if LVEF <55%. Class IIa AVR for rapid V_{max} progression (≥0.3 m/s/year), BNP ≥3×normal, or abnormal exercise test. BP drop >20 mm Hg triggers AVR consideration
Post-TAVR antithrombotic therapy	Aspirin monotherapy (class IIa), DAPT may be considered for 3-6 months (class IIb)	Aspirin monotherapy class I. No role for routine DAPT or anticoagulation unless another indication exists
Post-SAVR anticoagulation (bioprosthetic AVR)	Vitamin K antagonist for 3-6 months, then lifelong aspirin	Vitamin K antagonist or aspirin for 3 months, then lifelong aspirin

AVR=aortic valve replacement; BP=blood pressure; BNP=B-type natriuretic peptide; CT=computed tomography; DAPT=dual antiplatelet therapy; DSE=dobutamine stress echocardiography; HFpEF=heart failure with preserved ejection fraction; LVEF=left ventricular ejection fraction; SAVR=surgical aortic valve replacement; TAVR=transcatheter aortic valve replacement; V_{max} =peak aortic jet velocity

trials for their ability to provide complete cerebral coverage and reduce the risk of periprocedural stroke. Additionally, low profile transcatheter valve delivery systems designed to minimize vascular complications are being studied to enhance procedural safety and expand patient eligibility.^{175 176}

Collectively, these emerging technologies represent a promising frontier in the management of aortic stenosis. However, most remain in early clinical evaluation or have not yet received regulatory approval, and their long term safety and efficacy will need to be confirmed through rigorous controlled trials before being integrated into routine clinical practice.

Guidelines

Modern aortic stenosis management is anchored in guideline directed thresholds for intervention and a heart team framework for selecting the optimal procedure. In the US, the most recent society guidance remains the 2020 American College of Cardiology/American Heart Association (ACC/AHA) valvular heart disease guideline (no subsequent update). In Europe, the 2025 ESC/EACTS guidelines update and replace the 2021 ESC/EACTS document, with greater emphasis on multimodality imaging, structured heart valve centers, and lifetime management considerations (table 3).^{22 47 135 136}

The ACC/AHA and ESC/EACTS guidelines for aortic stenosis have evolved to prioritize earlier intervention and refine procedural selection between SAVR and

TAVR. Both recommend aortic valve replacement for symptomatic severe aortic stenosis (class I). In asymptomatic severe aortic stenosis, intervention is increasingly supported when high risk features are present such as left ventricular ejection fraction <50% (class I), very severe aortic stenosis ($V_{max} \geq 5.0$ -5.5 m/s, class IIa), abnormal exercise test results (class I if symptoms develop, class IIa for abnormal blood pressure response), rapid progression (≥0.3 m/s per year), or elevated biomarkers (B-type natriuretic peptide ≥3×normal, class IIa). The ESC/EACTS 2025 guidelines more explicitly support early intervention in selected asymptomatic patients: intervention should be considered in asymptomatic severe high gradient aortic stenosis with left ventricular ejection fraction ≥50% as an alternative to close surveillance when procedural risk is low (class IIa).

Procedure selection differs most clearly in the age based framework. The ACC/AHA 2020 guideline recommends SAVR in patients aged under 65 (or life expectancy >20 years), transfemoral TAVR in patients aged over 80 (or life expectancy <10 years), and shared decision making for those aged 65-80 without contraindication to transfemoral access. By contrast, ESC/EACTS 2025 recommends TAVR in patients aged 70 and over with tricuspid aortic stenosis and suitable anatomy (class Ia), SAVR in patients aged under 70 when surgical risk is low (class Ib), and individualized heart team decision making for all others, explicitly incorporating estimated life expectancy and lifetime management

(including feasibility of future coronary access and reintervention strategies).

Bicuspid aortic stenosis remains a surgical preference, especially in younger patients, though TAVR may be considered in older, high risk cases. In low flow, low gradient aortic stenosis, if severity is confirmed, aortic valve replacement is indicated (class I if contractile reserve is present, class IIa if absent but calcification is high). In paradoxical low flow, low gradient aortic stenosis—an entity still debated owing to its overlap with heart failure with preserved ejection fraction and moderate aortic stenosis—aortic valve replacement is recommended for symptomatic cases (ACC/AHA class I, ESC class IIa), reflecting differing thresholds. Current guidelines also recommend mechanical valves for patients aged <50-60 and bioprosthetic valves for those aged over 65 to avoid lifelong anticoagulation.^{16 22 47}

The 2021 National Institute for Health and Care Excellence (NICE) guidelines were the first comprehensive valvular disease guidance from the UK, incorporating cost effectiveness considerations.¹⁷⁷ While largely aligning with international standards, NICE took a more conservative stance on TAVR. For severe symptomatic aortic stenosis, it recommended SAVR for most low risk and intermediate risk patients, diverging from North American and European guidelines, which had expanded TAVR use. NICE explicitly noted that the cost of TAVR influenced this stance, marking a departure from purely clinical criteria.

Post-procedure antithrombotic strategies have also evolved. The European guidelines strongly favor aspirin alone after TAVR (class I), while American guidelines allow for short term dual antiplatelet therapy in select patients at low risk of bleeding (class IIb). Lifelong aspirin is recommended after bioprosthetic aortic valve replacement, with warfarin administered for a short period (3-6 months, class IIa) post-SAVR. In patients with long term anticoagulation needs, both guidelines recommend avoiding routine antiplatelet therapy to minimize the risk of bleeding. Overall, the guidelines emphasize early intervention for high risk asymptomatic aortic stenosis and advocate a tailored procedural approach based on individual risk profiles. The heart team remains central in optimizing decision making and long term outcomes.

Conclusion

Aortic stenosis management has shifted from a binary decision about valve replacement to a longitudinal, patient centered strategy spanning accurate diagnosis, optimal timing of intervention, and planning for decades of follow-up. Echocardiography remains the cornerstone for grading severity and assessing ventricular response, but contemporary practice increasingly depends on multimodality imaging including cardiac computed tomography to adjudicate discordant hemodynamics, define valve anatomy and calcific burden, and anticipate procedural challenges such as coronary access

and annular constraints. Randomized trials have established TAVR alongside surgery across the surgical risk spectrum, and the latest guideline frameworks reflect a more individualized approach that integrates age, anatomy, frailty, comorbidities, and life expectancy.

The next phase of the field is defined by “lifetime valve management,” especially as TAVR expands to younger and lower risk patients. Priorities include durable hemodynamics, prevention of prosthesis-patient mismatch in small annuli and valve in valve pathways, preservation of coronary access for future interventions, and mitigation of complications that can erode long term benefit—conduction disease and pacemaker dependency, valve thrombosis, stroke, and late structural valve deterioration. Key gaps remain—particularly in bicuspid aortic valve disease, which is under-represented in pivotal trials, and in defining optimal antithrombotic/anticoagulation

QUESTIONS FOR FUTURE RESEARCH

- In bicuspid aortic stenosis, does TAVR offer comparable long term outcomes to surgery, including durability, coronary access, and procedural complications?
- How do outcomes, procedural complexity, and patient safety compare between redo TAVR (TAVR in TAVR) and TAVR explantation, including coronary access and lifetime reintervention planning?
- What is the long term clinical impact of subclinical leaflet thrombosis detected by advanced imaging, and can targeted anticoagulation improve durability and outcomes?
- Could medical therapies (eg, SGLT2 inhibitors) or Lp(a) lipoprotein lowering strategies influence remodeling, heart failure outcomes, and/or calcification progression after valve intervention?
- Which frailty/comorbidity profiles predict limited survival or limited functional recovery after TAVR, and how should “survival with benefit” be operationalized in routine practice?

PATIENT INVOLVEMENT

We collaborated closely with patients by using structured surveys conducted in partnership with HeartValveSurgery.com, a patient led advocacy group dedicated to valvular heart disease education and patient empowerment. Patient surveys identified optimal management of bicuspid aortic stenosis in younger individuals and the comparative risks and benefits of redo TAVR versus surgical explantation for treatment of failure of index transcatheter valve as the most pressing concerns, prompting an in-depth evaluation of these topics in this review. Patient feedback emphasized the importance of understanding valve durability, long term outcomes, and strategies to preserve future treatment options, directly informing sections on structural valve degeneration and lifetime management.

strategies after TAVR across common scenarios (atrial fibrillation, recent percutaneous coronary intervention, suspected leaflet thrombosis). Progress will depend on long term randomized follow-up, high quality registries with complete ascertainment, and rigorous evaluation of next generation platforms and leaflet technologies to ensure that every decision is not just about replacing a valve but about safeguarding a patient's future.

Contributors: AG conceived the article, led the drafting and overall development of the manuscript, and supervised the work. SP performed the literature search, reviewed and synthesized the evidence, and contributed to drafting the manuscript. AG, SP, ML, JC, JJB, MAB, and RM helped interpret the evidence and revised the manuscript for important intellectual content. All authors approved the final version and agree to be accountable for the work. AG is the guarantor.

Funding sources: AG has received grant support from the National Heart, Lung, and Blood Institute (1R56HL175516-01).

Competing interests: AG reports grant support from National Heart Lung and Blood Institute and has received fees for consulting and speaking engagements with Boston Scientific and Edwards Lifesciences. ML reports institutional clinical research grants from Abbott, Boston Scientific, Edwards Lifesciences, and Medtronic. JC serves as editor in chief for *The Annals of Thoracic Surgery*. JJB has received speaker fees from Abbott Vascular. MAB discloses that his hospital receives speakers' honoraria and/or consulting fees on his behalf from Edwards Lifesciences, Medtronic, Abbott, and Artivion. RM reports grant support and consulting fees and/or honoraria from Boston Scientific, Edwards Lifesciences, Medtronic, Abbott, and Cordis; and grant support from Cedars-Sinai Medical Center. SP does not report any conflicts.

Provenance and peer review: Commissioned; externally peer reviewed.

- 1 Benfari G, Essayagh B, Michelena HI, et al. Severe aortic stenosis: secular trends of incidence and outcomes. *Eur Heart J* 2024;45:1877-86. doi:10.1093/eurheartj/ehad887.
- 2 Nkomo VT, Gardin JM, Skelton TN, Gottdiener JS, Scott CG, Enriquez-Sarano M. Burden of valvular heart diseases: a population-based study. *Lancet* 2006;368:1005-11. doi:10.1016/S0140-6736(06)69208-8.
- 3 Leon MB, Smith CR, Mack M, et al, PARTNER Trial Investigators. Transcatheter aortic-valve implantation for aortic stenosis in patients who cannot undergo surgery. *N Engl J Med* 2010;363:1597-607. doi:10.1056/NEJMoa1008232.
- 4 Smith CR, Leon MB, Mack MJ, et al, PARTNER Trial Investigators. Transcatheter versus surgical aortic-valve replacement in high-risk patients. *N Engl J Med* 2011;364:2187-98. doi:10.1056/NEJMoa1103510.
- 5 Makkar RR, Fontana GP, Jilaihawi H, et al, PARTNER Trial Investigators. Transcatheter aortic-valve replacement for inoperable severe aortic stenosis. *N Engl J Med* 2012;366:1696-704. doi:10.1056/NEJMoa1202277.
- 6 Leon MB, Smith CR, Mack MJ, et al, PARTNER 2 Investigators. Transcatheter or surgical aortic-valve replacement in intermediate-risk patients. *N Engl J Med* 2016;374:1609-20. doi:10.1056/NEJMoa1514616.
- 7 Makkar RR, Thourani VH, Mack MJ, et al, PARTNER 2 Investigators. Five-year outcomes of transcatheter or surgical aortic-valve replacement. *N Engl J Med* 2020;382:799-809. doi:10.1056/NEJMoa1910555.
- 8 Reardon MJ, Van Mieghem NM, Popma JJ, et al, SURTAVI Investigators. Surgical or transcatheter aortic-valve replacement in intermediate-risk patients. *N Engl J Med* 2017;376:1321-31. doi:10.1056/NEJMoa1700456.
- 9 Kapadia SR, Leon MB, Makkar RR, et al, PARTNER trial investigators. 5-year outcomes of transcatheter aortic valve replacement compared with standard treatment for patients with inoperable aortic stenosis (PARTNER 1): a randomised controlled trial. *Lancet* 2015;385:2485-91. doi:10.1016/S0140-6736(15)60290-2.
- 10 Mack MJ, Leon MB, Smith CR, et al, PARTNER 1 trial investigators. 5-year outcomes of transcatheter aortic valve replacement or surgical aortic valve replacement for high surgical risk patients with aortic stenosis (PARTNER 1): a randomised controlled trial. *Lancet* 2015;385:2477-84. doi:10.1016/S0140-6736(15)60308-7.
- 11 Mack MJ, Leon MB, Thourani VH, et al, PARTNER 3 Investigators. Transcatheter aortic-valve replacement with a balloon-expandable valve in low-risk patients. *N Engl J Med* 2019;380:1695-705. doi:10.1056/NEJMoa1814052.
- 12 Popma JJ, Deeb GM, Yakubov SJ, et al, Evolut Low Risk Trial Investigators. Transcatheter aortic-valve replacement with a self-expanding valve in low-risk patients. *N Engl J Med* 2019;380:1706-15. doi:10.1056/NEJMoa1816885.
- 13 Mack MJ, Leon MB, Thourani VH, et al, PARTNER 3 Investigators. Transcatheter aortic-valve replacement in low-risk patients at five years. *N Engl J Med* 2023;389:1949-60. doi:10.1056/NEJMoa2307447.
- 14 Kodali SK, Williams MR, Smith CR, et al, PARTNER Trial Investigators. Two-year outcomes after transcatheter or surgical aortic-valve replacement. *N Engl J Med* 2012;366:1686-95. doi:10.1056/NEJMoa1200384.
- 15 Adams DH, Popma JJ, Reardon MJ, et al, U.S. CoreValve Clinical Investigators. Transcatheter aortic-valve replacement with a self-expanding prosthesis. *N Engl J Med* 2014;370:1790-8. doi:10.1056/NEJMoa1400590.
- 16 Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA Guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 2021;143:e35-71. doi:10.1161/CIR.0000000000000932.
- 17 Clavel MA, Burwash IG, Pibarot P. Cardiac imaging for assessing low-gradient severe aortic stenosis. *JACC Cardiovasc Imaging* 2017;10:185-202. doi:10.1016/j.jcmg.2017.01.002.
- 18 Park J, Kim J, Jeon J, et al. Artificial intelligence-enhanced comprehensive assessment of the aortic valve stenosis continuum in echocardiography. *EBioMedicine* 2025;112:105560. doi:10.1016/j.ebiom.2025.105560.
- 19 Fortuni F, Ciliberti G, De Chiara B, et al. Advancements and applications of artificial intelligence in cardiovascular imaging: a comprehensive review. *Eur Heart J Imaging Methods Pract* 2024;2:qyae136. doi:10.1093/ehjimp/qyae136.
- 20 Sengupta PP, Klain J, Lee S-P, Oh JK, Smits ALPM. The future of valvular heart disease assessment and therapy. *Lancet* 2024;403:1590-602. doi:10.1016/S0140-6736(23)02754-X.
- 21 Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA Guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 2021;143:e72-227. doi:10.1161/CIR.0000000000000923.
- 22 Vahanian A, Beyersdorf F, Praz F, et al, ESC/EACTS Scientific Document Group. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J* 2022;43:561-632. doi:10.1093/eurheartj/ehab395.
- 23 Søndergaard L, Ihlemann N, Capodanno D, et al. Durability of transcatheter and surgical bioprosthetic aortic valves in patients at lower surgical risk. *J Am Coll Cardiol* 2019;73:546-53. doi:10.1016/j.jacc.2018.10.083.
- 24 Modine T, Tchéché D, Van Mieghem NM, et al. Three-year outcomes following TAVR in younger (<75 years) low-surgical-risk severe aortic stenosis patients. *Circ Cardiovasc Interv* 2024;17:e014018. doi:10.1161/CIRCINTERVENTIONS.124.014018.
- 25 Eveborn GW, Schirmer H, Heggelund G, Lunde P, Rasmussen K. The evolving epidemiology of valvular aortic stenosis. The Tromsø study. *Heart* 2013;99:396-400. doi:10.1136/heartjnl-2012-302265.
- 26 Osnabrügge RL, Mylotte D, Head SJ, et al. Aortic stenosis in the elderly: disease prevalence and number of candidates for transcatheter aortic valve replacement: a meta-analysis and modeling study. *J Am Coll Cardiol* 2013;62:1002-12. doi:10.1016/j.jacc.2013.05.015.
- 27 Patel DK, Green KD, Fudim M, Harrell FE, Wang TJ, Robbins MA. Racial differences in the prevalence of severe aortic stenosis. *J Am Heart Assoc* 2014;3:e000879. doi:10.1161/JAHA.114.000879.
- 28 Gupta A, Mori M, Wang Y, et al. Racial/ethnic disparities in aortic valve replacement among Medicare beneficiaries in the United States, 2012-2019. *Am J Med* 2024;137:321-30. doi:10.1016/j.amjmed.2023.12.026.
- 29 Lindman BR, Clavel M-A, Mathieu P, et al. Calcific aortic stenosis. *Nat Rev Dis Primers* 2016;2:16006. doi:10.1038/nrdp.2016.6.
- 30 Dweck MR, Boon NA, Newby DE. Calcific aortic stenosis: a disease of the valve and the myocardium. *J Am Coll Cardiol* 2012;60:1854-63. doi:10.1016/j.jacc.2012.02.093.
- 31 Otto CM. *Calcific aortic stenosis—time to look more closely at the valve*. Mass Medical Soc, 2008:1395-8.
- 32 Praz F, Beyersdorf F, Haugaa K, Prendergast B. Valvular heart disease: from mechanisms to management. *Lancet* 2024;403:1576-89. doi:10.1016/S0140-6736(23)02755-1.
- 33 Thanassoulis G, Campbell CY, Owens DS, et al, CHARGE Extracoronary Calcium Working Group. Genetic associations

- with valvular calcification and aortic stenosis. *N Engl J Med* 2013;368:503-12. doi:10.1056/NEJMoa1109034
- 34 Arsenault BJ, Loganath K, Girard A, et al. Lipoprotein(a) and calcific aortic valve stenosis progression: a systematic review and meta-analysis. *JAMA Cardiol* 2024;9:835-42. doi:10.1001/jamacardio.2024.1882
 - 35 Lindman BR, Sukul D, Dweck MR, et al. Evaluating medical therapy for calcific aortic stenosis: JACC state-of-the-art review. *J Am Coll Cardiol* 2021;78:2354-76. doi:10.1016/j.jacc.2021.09.1367
 - 36 Raposeiras-Roubin S, Amat-Santos IJ, Rossello X, et al, DapaTAVI Investigators. Dapagliflozin in patients undergoing transcatheter aortic-valve implantation. *N Engl J Med* 2025;392:1396-405. doi:10.1056/NEJMoa2500366
 - 37 Tzemos N, Therrien J, Yip J, et al. Outcomes in adults with bicuspid aortic valves. *JAMA* 2008;300:1317-25. doi:10.1001/jama.300.11.1317
 - 38 Sillesen AS, Vøgg O, Pihl C, et al. Prevalence of bicuspid aortic valve and associated aortopathy in newborns in Copenhagen, Denmark. *JAMA* 2021;325:561-7. doi:10.1001/jama.2020.27205
 - 39 Verma S, Siu SC. Aortic dilatation in patients with bicuspid aortic valve. *N Engl J Med* 2014;370:1920-9. doi:10.1056/NEJMra1207059
 - 40 Borger MA, Fedak PWM, Stephens EH, et al. The American Association for Thoracic Surgery consensus guidelines on bicuspid aortic valve-related aortopathy: Full online-only version. *J Thorac Cardiovasc Surg* 2018;156:e41-74. doi:10.1016/j.jtcvs.2018.02.115
 - 41 Makkar RR, Yoon S-H, Chakravarty T, et al. Association between transcatheter aortic valve replacement for bicuspid vs tricuspid aortic stenosis and mortality or stroke among patients at low surgical risk. *JAMA* 2021;326:1034-44. doi:10.1001/jama.2021.13346
 - 42 Marijon E, Mirabel M, Celermajer DS, Jouven X. Rheumatic heart disease. *Lancet* 2012;379:953-64. doi:10.1016/S0140-6736(11)61171-9
 - 43 Garg V, Muth AN, Ransom JF, et al. Mutations in NOTCH1 cause aortic valve disease. *Nature* 2005;437:270-4. doi:10.1038/nature03940
 - 44 Probst V, Le Scouarnec S, Legendre A, et al. Familial aggregation of calcific aortic valve stenosis in the western part of France. *Circulation* 2006;113:856-60. doi:10.1161/CIRCULATIONAHA.105.569467
 - 45 Dweck MR, Loganath K, Bing R, et al. Multi-modality imaging in aortic stenosis: an EACVI clinical consensus document. *Eur Heart J Cardiovasc Imaging* 2023;24:1430-43. doi:10.1093/ehjci/jead153
 - 46 Baumgartner H, Hung J, Bermejo J, et al. Recommendations on the echocardiographic assessment of aortic valve stenosis: a focused update from the European Association of Cardiovascular Imaging and the American Society of Echocardiography. *J Am Soc Echocardiogr* 2017;30:372-92. doi:10.1016/j.echo.2017.02.009
 - 47 Praz F, Borger MA, Lanz J, et al, ESC/EACTS Scientific Document Group. 2025 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J* 2025;46:4635-736. doi:10.1093/eurheartj/ehaf194
 - 48 Treibel TA, López B, González A, et al. Reappraising myocardial fibrosis in severe aortic stenosis: an invasive and non-invasive study in 133 patients. *Eur Heart J* 2018;39:699-709. doi:10.1093/eurheartj/ehx353
 - 49 Park SJ, Cho SW, Kim SM, et al. Assessment of myocardial fibrosis using multimodality imaging in severe aortic stenosis: comparison with histologic fibrosis. *JACC Cardiovasc Imaging* 2019;12:109-19. doi:10.1016/j.jcmg.2018.05.028
 - 50 Vollema EM, Sugimoto T, Shen M, et al. Association of left ventricular global longitudinal strain with asymptomatic severe aortic stenosis: natural course and prognostic value. *JAMA Cardiol* 2018;3:839-47. doi:10.1001/jamacardio.2018.2288
 - 51 Pawade T, Sheth T, Guzzetti E, Dweck MR, Clavel M-A. Why and how to measure aortic valve calcification in patients with aortic stenosis. *JACC Cardiovasc Imaging* 2019;12:1835-48. doi:10.1016/j.jcmg.2019.01.045
 - 52 Lee G, Chikwe J, Milojevic M, et al. ESC/EACTS vs. ACC/AHA guidelines for the management of severe aortic stenosis. *Eur Heart J* 2023;44:796-812. doi:10.1093/eurheartj/ehac803
 - 53 Cartlidge TR, Bing R, Kwiecinski J, et al. Contrast-enhanced computed tomography assessment of aortic stenosis. *Heart* 2021;107:1905-11. doi:10.1136/heartjnl-2020-318556
 - 54 Okada A, Beckmann E, Rocher EE, et al. Preprocedural computed tomography planning for surgical aortic valve replacement. *Ann Thorac Surg* 2024;117:1154-62. doi:10.1016/j.athoracsurg.2024.02.017
 - 55 Musa TA, Treibel TA, Vassiliou VS, et al. Myocardial scar and mortality in severe aortic stenosis: data from the BSCMR valve consortium. *Circulation* 2018;138:1935-47. doi:10.1161/CIRCULATIONAHA.117.032839
 - 56 Loganath K, Craig NJ, Everrett RJ, et al, EVOLVED investigators. Early intervention in patients with asymptomatic severe aortic stenosis and myocardial fibrosis: the EVOLVED randomized clinical trial. *JAMA* 2025;333:213-21. doi:10.1001/jama.2024.22730
 - 57 Généreux P, Pibarot P, Redfors B, et al. Staging classification of aortic stenosis based on the extent of cardiac damage. *Eur Heart J* 2017;38:3351-8. doi:10.1093/eurheartj/ehx381
 - 58 Abdelfattah OM, Jacquemyn X, Sá MP, et al. Cardiac damage staging predicts outcomes in aortic valve stenosis after aortic valve replacement: meta-analysis. *JACC Adv* 2024;3:100959. doi:10.1016/j.jaccadv.2024.100959
 - 59 Snir AD, Ng MK, Strange G, Playford D, Stewart S, Celermajer DS, National Echo Database of Australia. Cardiac damage staging classification predicts prognosis in all the major subtypes of severe aortic stenosis: insights from the National Echo Database Australia. *J Am Soc Echocardiogr* 2021;34:1137-1147.e13. doi:10.1016/j.echo.2021.05.017
 - 60 Leclercq F, Robert P, Akodad M, et al. Prior balloon valvuloplasty versus direct transcatheter aortic valve replacement: results from the DIRECTAVI trial. *JACC Cardiovasc Interv* 2020;13:594-602. doi:10.1016/j.jcin.2019.12.006
 - 61 Hammermeister KE, Sethi GK, Henderson WG, Oprian C, Kim T, Rahimtoola S, Veterans Affairs Cooperative Study on Valvular Heart Disease. A comparison of outcomes in men 11 years after heart-valve replacement with a mechanical valve or bioprosthesis. *N Engl J Med* 1993;328:1289-96. doi:10.1056/NEJM199305063281801
 - 62 Hammermeister K, Sethi GK, Henderson WG, Grover FL, Oprian C, Rahimtoola SH. Outcomes 15 years after valve replacement with a mechanical versus a bioprosthetic valve: final report of the Veterans Affairs randomized trial. *J Am Coll Cardiol* 2000;36:1152-8. doi:10.1016/S0735-1097(00)00834-2
 - 63 Goldstone AB, Chiu P, Baiocchi M, et al. Mechanical or biologic prostheses for aortic-valve and mitral-valve replacement. *N Engl J Med* 2017;377:1847-57. doi:10.1056/NEJMoa1613792
 - 64 Bowdish ME, Mehaffey JH, Chang S-C, et al. Bioprosthetic vs mechanical aortic valve replacement in patients 40 to 75 years of age. *J Am Coll Cardiol* 2025;85:1289-98. doi:10.1016/j.jacc.2025.01.013
 - 65 Mazine A, El-Hamamsy I, Verma S, et al. Ross procedure in adults for cardiologists and cardiac surgeons: JACC state-of-the-art review. *J Am Coll Cardiol* 2018;72:2761-77. doi:10.1016/j.jacc.2018.08.2200
 - 66 Notenboom ML, Melina G, Veen KM, et al. Long-term clinical and echocardiographic outcomes following the Ross procedure: a post hoc analysis of a randomized clinical trial. *JAMA Cardiol* 2024;9:6-14. doi:10.1001/jamacardio.2023.4090
 - 67 Mazine A, Rocha RV, El-Hamamsy I, et al. Ross procedure vs mechanical aortic valve replacement in adults: a systematic review and meta-analysis. *JAMA Cardiol* 2018;3:978-87. doi:10.1001/jamacardio.2018.2946
 - 68 Buratto E, Shi WY, Wynne R, et al. Improved survival after the Ross procedure compared with mechanical aortic valve replacement. *J Am Coll Cardiol* 2018;71:1337-44. doi:10.1016/j.jacc.2018.01.048
 - 69 Sharabiani MT, Dorobantu DM, Mahani AS, et al. Aortic valve replacement and the Ross operation in children and young adults. *J Am Coll Cardiol* 2016;67:2858-70. doi:10.1016/j.jacc.2016.04.021
 - 70 Manouguian S, Seybold-Epting W. Patch enlargement of the aortic valve ring by extending the aortic incision into the anterior mitral leaflet. New operative technique. *J Thorac Cardiovasc Surg* 1979;78:402-12. doi:10.1016/S0022-5223(19)38105-X
 - 71 Nicks R, Cartmill T, Bernstein L. Hypoplasia of the aortic root. The problem of aortic valve replacement. *Thorax* 1970;25:339-46. doi:10.1136/thx.25.3.339
 - 72 Baumbach A, van Royen N, Amat-Santos IJ, et al, LANDMARK trial investigators. LANDMARK comparison of early outcomes of newer-generation Myval transcatheter heart valve series with contemporary valves (Sapien and Evolut) in real-world individuals with severe symptomatic native aortic stenosis: a randomised non-inferiority trial. *Lancet* 2024;403:2695-708. doi:10.1016/S0140-6736(24)00821-3
 - 73 Junquera L, Kalavrouziotis D, Dumont E, Rodés-Cabau J, Mohammadi S. Paradigm shifts in alternative access for transcatheter aortic valve replacement: An update. *J Thorac Cardiovasc Surg* 2023;165:1359-1370.e2. doi:10.1016/j.jtcvs.2021.04.075
 - 74 Dawadi S, Oli PR, Shrestha DB, et al. Transcatheter versus trans-axillary/subclavian transcatheter aortic valve replacement (TAVR): A systematic review and meta-analysis. *Curr Probl Cardiol* 2024;49:102488. doi:10.1016/j.cpcardiol.2024.102488
 - 75 Makkar RR, Cheng W, Waksman R, et al. Self-expanding intra-annular versus commercially available transcatheter heart valves in high and extreme risk patients with severe aortic stenosis (PORTICO IDE): a randomised, controlled, non-inferiority trial. *Lancet* 2020;396:669-83. doi:10.1016/S0140-6736(20)31358-1
 - 76 Tamburino C, Bleiziffer S, Thiele H, et al. Comparison of self-expanding bioprostheses for transcatheter aortic valve replacement in patients with symptomatic severe aortic stenosis: SCOPE

- 2 randomized clinical trial. *Circulation* 2020;142:2431-42. doi:10.1161/CIRCULATIONAHA.120.051547
- 77 Lanz J, Kim W-K, Walther T, et al, SCOPE I investigators. Safety and efficacy of a self-expanding versus a balloon-expandable bioprosthesis for transcatheter aortic valve replacement in patients with symptomatic severe aortic stenosis: a randomised non-inferiority trial. *Lancet* 2019;394:1619-28. doi:10.1016/S0140-6736(19)32220-2
 - 78 Herrmann HC, Mehrran R, Blackman DJ, et al, SMART Trial Investigators. Self-expanding or balloon-expandable TAVR in patients with a small aortic annulus. *N Engl J Med* 2024;390:1959-71. doi:10.1056/NEJMoa2312573
 - 79 Abdel-Wahab M, Mehilli J, Frerker C, et al, CHOICE investigators. Comparison of balloon-expandable vs self-expandable valves in patients undergoing transcatheter aortic valve replacement: the CHOICE randomized clinical trial. *JAMA* 2014;311:1503-14. doi:10.1001/jama.2014.3316
 - 80 Thiele H, Kurz T, Feistritz HJ, et al. Comparison of newer generation self-expandable vs. balloon-expandable valves in transcatheter aortic valve implantation: the randomized SOLVE-TAVI trial. *Eur Heart J* 2020;41:1890-9. doi:10.1093/eurheartj/ehaa036
 - 81 Vlastra W, Chandrasekhar J, Muñoz-García AJ, et al. Comparison of balloon-expandable vs. self-expandable valves in patients undergoing transfemoral transcatheter aortic valve implantation: from the CENTER-collaboration. *Eur Heart J* 2019;40:456-65. doi:10.1093/eurheartj/ehy805
 - 82 Leon MB, Mack MJ, Pibarot P, et al. Transcatheter or surgical aortic valve replacement in low-risk patients at 7 years. *N Engl J Med* 2025. doi:10.1056/NEJMoa2509766
 - 83 Forrest JK, Yakubov SJ, Deeb GM, et al, Low Risk Trial Investigators. 5-year outcomes after transcatheter or surgical aortic valve replacement in low-risk patients with aortic stenosis. *J Am Coll Cardiol* 2025;85:1523-32. doi:10.1016/j.jacc.2025.03.004
 - 84 Forrest JK, Deeb GM, Yakubov SJ, et al, Evolut Low Risk Trial Investigators. 4-year outcomes of patients with aortic stenosis in the Evolut low risk trial. *J Am Coll Cardiol* 2023;82:2163-5. doi:10.1016/j.jacc.2023.09.813
 - 85 Forrest JK, Yakubov SJ, Deeb GM, Reardon MJ, Evolut Low Risk Trial Investigators. Six-year outcomes after transcatheter vs surgical aortic valve replacement in low-risk patients with aortic stenosis. *J Am Coll Cardiol* 2026;16:05423-9. doi:10.1016/j.jacc.2026.02.0563
 - 86 Gupta A, Cohen DJ. When the curve bends: low-risk TAVR beyond year 5. *J Am Coll Cardiol* 2026;26:05436-7. doi:10.1016/j.jacc.2026.02.0566
 - 87 UK TAVI Trial Investigators. Effect of transcatheter aortic valve implantation vs surgical aortic valve replacement on all-cause mortality in patients with aortic stenosis: a randomized clinical trial. *JAMA* 2022;327:1875-87. doi:10.1001/jama.2022.5776
 - 88 Tchetché D, Pibarot P, Bax JJ, et al. Transcatheter vs. surgical aortic valve replacement in women: the RHEIA trial. *Eur Heart J* 2025;46:2079-88. doi:10.1093/eurheartj/ehaf133
 - 89 Eltchaninoff H, Bonaros N, Prendergast B, et al. Rationale and design of a prospective, randomized, controlled, multicenter study to evaluate the safety and efficacy of transcatheter heart valve replacement in female patients with severe symptomatic aortic stenosis requiring aortic valve intervention (Randomized research in womEn all comers with Aortic stenosis [RHEIA] trial). *Am Heart J* 2020;228:27-35. doi:10.1016/j.ahj.2020.06.016
 - 90 Blankenberg S, Seiffert M, Vonthein R, et al, DEDICATE-DZHK6 Trial Investigators. Transcatheter or surgical treatment of aortic valve stenosis. *N Engl J Med* 2024;390:1572-83. doi:10.1056/NEJMoa2400685
 - 91 Makkar RR, Ramana RK, Gnall E, et al, ACURATE IDE study investigators. ACURATE neo2 valve versus commercially available transcatheter heart valves in patients with severe aortic stenosis (ACURATE IDE): a multicentre, randomised, controlled, non-inferiority trial. *Lancet* 2025;405:2061-74. doi:10.1016/S0140-6736(25)00319-8
 - 92 Serruys PW, Tobe A, van Royen N, et al. 1-year outcomes of novel balloon-expandable versus contemporary transcatheter heart valves in severe aortic stenosis: the LANDMARK trial. *J Am Coll Cardiol* 2026;87:362-81. doi:10.1016/j.jacc.2025.10.076
 - 93 Hoffman JL, Kaplan S. The incidence of congenital heart disease. *J Am Coll Cardiol* 2002;39:1890-900. doi:10.1016/S0735-1097(02)01886-7
 - 94 Roberts WC, Ko JM. Frequency by decades of unicuspid, bicuspid, and tricuspid aortic valves in adults having isolated aortic valve replacement for aortic stenosis, with or without associated aortic regurgitation. *Circulation* 2005;111:920-5. doi:10.1161/01.CIR.0000155623.48408.C5
 - 95 Forrest JK, Kaple RK, Ramlawi B, et al. Transcatheter aortic valve replacement in bicuspid versus tricuspid aortic valves from the STS/ACC TVT registry. *JACC Cardiovasc Interv* 2020;13:1749-59. doi:10.1016/j.jcin.2020.03.022
 - 96 Makkar RR, Yoon SH, Leon MB, et al. Association between transcatheter aortic valve replacement for bicuspid vs tricuspid aortic stenosis and mortality or stroke. *JAMA* 2019;321:2193-202. doi:10.1001/jama.2019.7108
 - 97 Williams MR, Jilalawi H, Makkar R, et al. The PARTNER 3 bicuspid registry for transcatheter aortic valve replacement in low-surgical-risk patients. *JACC Cardiovasc Interv* 2022;15:523-32. doi:10.1016/j.jcin.2022.01.279
 - 98 Forrest JK, Ramlawi B, Deeb GM, et al. Transcatheter aortic valve replacement in low-risk patients with bicuspid aortic valve stenosis. *JAMA Cardiol* 2021;6:50-7. doi:10.1001/jamacardio.2020.4738
 - 99 Yoon SH, Bleiziffer S, De Backer O, et al. Outcomes in transcatheter aortic valve replacement for bicuspid versus tricuspid aortic valve stenosis. *J Am Coll Cardiol* 2017;69:2579-89. doi:10.1016/j.jacc.2017.03.017
 - 100 Deeb GM, Reardon MJ, Ramlawi B, et al. Propensity-matched 1-year outcomes following transcatheter aortic valve replacement in low-risk bicuspid and tricuspid patients. *JACC Cardiovasc Interv* 2022;15:511-22. doi:10.1016/j.jcin.2021.10.027
 - 101 Moscarelli M, Venkatesh A, Berland K, et al. Post-implant transcatheter aortic prosthesis deformation: tricuspid versus bicuspid valve. *Eur J Cardiothorac Surg* 2024;67:ezae451. doi:10.1093/ejcts/ezae451
 - 102 Nagasaka T, Patel V, Shechter A, et al. Impact of balloon-expandable TAVR valve deformation and calcium distribution on outcomes in bicuspid aortic valve. *JACC Cardiovasc Interv* 2024;17:2023-37. doi:10.1016/j.jcin.2024.07.018
 - 103 Yoon SH, Kim WK, Dhoble A, et al, Bicuspid Aortic Valve Stenosis Transcatheter Aortic Valve Replacement Registry Investigators. Bicuspid aortic valve morphology and outcomes after transcatheter aortic valve replacement. *J Am Coll Cardiol* 2020;76:1018-30. doi:10.1016/j.jacc.2020.07.005
 - 104 Jørgensen TH, Thyregod GH, Savontaus M, et al, NOTION-2 investigators. Transcatheter aortic valve implantation in low-risk tricuspid or bicuspid aortic stenosis: the NOTION-2 trial. *Eur Heart J* 2024;45:3804-14. doi:10.1093/eurheartj/ehae331
 - 105 Samim D, Siepe M, Jüni P, et al. Transcatheter treatment of bicuspid aortic valve stenosis: from observational studies to randomized clinical trials. *Structural Heart* 2025;9:100754. doi:10.1016/j.shj.2025.100754
 - 106 Généreux P, Schwartz P, Oldemeyer JB, et al, EARLY TAVR Trial Investigators. Transcatheter aortic valve replacement for asymptomatic severe aortic stenosis. *N Engl J Med* 2025;392:217-27. doi:10.1056/NEJMoa2405880
 - 107 Kang D-H, Park S-J, Lee S-A, et al. Early surgery or conservative care for asymptomatic aortic stenosis. *N Engl J Med* 2020;382:111-9. doi:10.1056/NEJMoa1912846
 - 108 Banovic M, Putnik S, Penicka M, et al, AVATAR Trial Investigators*. Aortic valve replacement versus conservative treatment in asymptomatic severe aortic stenosis: the AVATAR trial. *Circulation* 2022;145:648-58. doi:10.1161/CIRCULATIONAHA.121.057639
 - 109 Banovic M, Putnik S, Da Costa BR, et al. Aortic valve replacement vs. conservative treatment in asymptomatic severe aortic stenosis: long-term follow-up of the AVATAR trial. *Eur Heart J* 2024;45:4526-35. doi:10.1093/eurheartj/ehae585
 - 110 Van Mieghem NM, Elmariah S, Spitzer E, et al. Transcatheter aortic valve replacement in patients with systolic heart failure and moderate aortic stenosis. *JACC* 2025;85:878-90. doi:10.1016/j.jacc.2024.10.070
 - 111 Généreux P, Gupta A, Makkar RR. TAVR in patients with moderate aortic stenosis and heart failure. *JACC* 2025;85:891-3. doi:10.1016/j.jacc.2024.10.088
 - 112 Makkar RR, Fontana G, Jilalawi H, et al. Possible subclinical leaflet thrombosis in bioprosthetic aortic valves. *N Engl J Med* 2015;373:2015-24. doi:10.1056/NEJMoa1509233
 - 113 Chakravarty T, Søndergaard L, Friedman J, et al, RESOLVESAVORY Investigators. Subclinical leaflet thrombosis in surgical and transcatheter bioprosthetic aortic valves: an observational study. *Lancet* 2017;389:2383-92. doi:10.1016/S0140-6736(17)30757-2
 - 114 Généreux P, Piazza N, Alu MC, et al, VARC-3 WRITING COMMITTEE. Valve Academic Research Consortium 3: updated endpoint definitions for aortic valve clinical research. *Eur Heart J* 2021;42:1825-57. doi:10.1093/eurheartj/ehaa799
 - 115 Jilalawi H, Asch FM, Manasse E, et al. Systematic CT methodology for the evaluation of subclinical leaflet thrombosis. *JACC Cardiovasc Imaging* 2017;10:461-70. doi:10.1016/j.jcmg.2017.02.005
 - 116 Fukui M, Bapat VN, Garcia S, et al. Deformation of transcatheter aortic valve prostheses: implications for hypoattenuating leaflet thickening and clinical outcomes. *Circulation* 2021;121:058339. doi:10.1161/CIRCULATIONAHA.121.058339
 - 117 Hein M, Schoechlin S, Schulz U, et al. Long-term follow-up of hypoattenuated leaflet thickening after transcatheter aortic

- valve replacement. *JACC Cardiovasc Interv* 2022;15:1113-22. doi:10.1016/j.jcin.2022.04.018.
- 118 Yanagisawa R, Tanaka M, Yashima F, et al. Early and late leaflet thrombosis after transcatheter aortic valve replacement: a multicenter initiative from the OCEAN-TAVI Registry. *Circ Cardiovasc Interv* 2019;12:e007349. doi:10.1161/CIRCINTERVENTIONS.118.007349
 - 119 Dangas GD, Tijssen JGP, Wöhrle J, et al, GALILEO Investigators. A controlled trial of rivaroxaban after transcatheter aortic-valve replacement. *N Engl J Med* 2020;382:120-9. doi:10.1056/NEJMoa19111425
 - 120 Collet JP, Van Belle E, Thiele H, et al, ATLANTIS Investigators of the ACTION Group. Apixaban vs. standard of care after transcatheter aortic valve implantation: the ATLANTIS trial. *Eur Heart J* 2022;43:2783-97. doi:10.1093/eurheartj/ehac242.
 - 121 Al-Kassou B, Kandt J, Lohde L, et al. Safety and efficacy of protamine administration for prevention of bleeding complications in patients undergoing TAVR. *JACC Cardiovasc Interv* 2020;13:1471-80. doi:10.1016/j.jcin.2020.03.041.
 - 122 Al-Kassou B, Veulemans V, Shamekhi J, et al. Optimal protamine-to-heparin dosing ratio for the prevention of bleeding complications in patients undergoing TAVR-A multicenter experience. *Clin Cardiol* 2023;46:67-75. doi:10.1002/clc.23936.
 - 123 Huded CP, Tuzcu EM, Krishnaswamy A, et al. Association between transcatheter aortic valve replacement and early postprocedural stroke. *JAMA* 2019;321:2306-15. doi:10.1001/jama.2019.7525
 - 124 Carroll JD, Mack MJ, Vemulapalli S, et al. STS-ACC TVT registry of transcatheter aortic valve replacement. *J Am Coll Cardiol* 2020;76:2492-516. doi:10.1016/j.jacc.2020.09.595
 - 125 Kapadia SR, Makkak R, Leon M, et al, PROTECTED TAVR Investigators. Cerebral embolic protection during transcatheter aortic-valve replacement. *N Engl J Med* 2022;387:1253-63. doi:10.1056/NEJMoa2204961.
 - 126 Makkak RR, Gupta A, Waggoner TE, et al. Cerebral embolic protection by geographic region: a post hoc analysis of the PROTECTED TAVR randomized clinical trial. *JAMA Cardiol* 2025;10:17-24. doi:10.1001/jamacardio.2024.4278
 - 127 Butala NM, Kapadia SR, Secemsky EA, et al. Impact of cerebral embolic protection devices on disabling stroke after TAVR: updated results from the STS/ACC TVT registry. *Circ Cardiovasc Interv* 2024;17:e013697. doi:10.1161/CIRCINTERVENTIONS.123.013697
 - 128 Lowenstern A, Hung A, Manandhar P, et al. Association of transcatheter aortic valve replacement reimbursement, new technology add-on payment, and procedure volumes with embolic protection device use. *JAMA Cardiol* 2022;7:945-52. doi:10.1001/jamacardio.2022.2608.
 - 129 Saad M, Has P, Elkaryoni A, Alkhoul M. Embolic protection device use with transcatheter aortic valve replacement after PROTECTED TAVR trial. *JACC Cardiovasc Interv* 2025;18:1209-10. doi:10.1016/j.jcin.2025.01.445.
 - 130 Kharbanda RK, Perkins AD, Kennedy J, et al. Routine cerebral embolic protection in transcatheter aortic valve implantation: rationale and design of the randomised British Heart Foundation PROTECT-TAVI trial. *EuroIntervention* 2023;18:1428-35. doi:10.4244/EIJ-D-22-00713
 - 131 Kharbanda RK, Kennedy J, Jamal Z, et al, BHF PROTECT-TAVI Investigators. Routine cerebral embolic protection during transcatheter aortic-valve implantation. *N Engl J Med* 2025;392:2403-12. doi:10.1056/NEJMoa2415120.
 - 132 Shekhar S, Isogai T, Agrawal A, et al. Outcomes and predictors of stroke after transcatheter aortic valve replacement in the cerebral protection device era. *J Am Heart Assoc* 2024;13:e034298. doi:10.1161/JAHA.124.034298
 - 133 Windecker S, Okuno T, Unbehauen A, Mack M, Kapadia S, Falk V. Which patients with aortic stenosis should be referred to surgery rather than transcatheter aortic valve implantation? *Eur Heart J* 2022;43:2729-50. doi:10.1093/eurheartj/ehac105
 - 134 Kedhi E, Hermanides RS, Dambrink JE, et al, TCW study group. TransCatheter aortic valve implantation and fractional flow reserve-guided percutaneous coronary intervention versus conventional surgical aortic valve replacement and coronary bypass grafting for treatment of patients with aortic valve stenosis and complex or multivessel coronary disease (TCW): an international, multicentre, prospective, open-label, non-inferiority, randomised controlled trial. *Lancet* 2024;404:2593-602. doi:10.1016/S0140-6736(24)02100-7.
 - 135 Otto CM, Nishimura RA, Bonow RO, et al, Writing Committee Members. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol* 2021;77:450-500. doi:10.1016/j.jacc.2020.11.035
 - 136 Coisne A, Lancellotti P, Habib G, et al, EuroValve Consortium. ACC/AHA and ESC/EACTS guidelines for the management of valvular heart diseases: JACC guideline comparison. *J Am Coll Cardiol* 2023;82:721-34. doi:10.1016/j.jacc.2023.05.061
 - 137 Lønborg J, Jabbari R, Sabbah M, et al, NOTION-3 Study Group. PCI in patients undergoing transcatheter aortic-valve implantation. *N Engl J Med* 2024;391:2189-200. doi:10.1056/NEJMoa2401513.
 - 138 ClinicalTrials.gov. Optimal timing of transcatheter aortic valve implantation and percutaneous coronary intervention - The TAVI PCI Trial (TAVI-PCI) NCT04310046. Updated 2023 Dec 21. <https://clinicaltrials.gov/study/NCT04310046?term=NCT04310046&rank=1>
 - 139 Dhoble A, Ahmed T, McKay RG, et al. Timing and outcomes of PCI in conjunction with TAVR with balloon-expandable valves. *JACC Cardiovasc Interv* 2025;18:244-54. doi:10.1016/j.jcin.2024.10.055.
 - 140 ClinicalTrials.gov. Staged complete revascularization for coronary artery disease vs medical management alone in patients with AS undergoing transcatheter aortic valve replacement (COMPLETE TAVR). Updated 2025. <https://clinicaltrials.gov/study/NCT04634240>
 - 141 ClinicalTrials.gov. Comparative effectiveness study of staged complete revascularization with percutaneous coronary intervention to treat coronary artery disease vs medical management alone in patients with symptomatic aortic valve stenosis undergoing elective transfemoral transcatheter aortic valve replacement: The COMPLETE TAVR Study. National Library of Medicine (US). <https://clinicaltrials.gov/study/NCT04634240>
 - 142 Pibarot P, Herrmann HC, Wu C, et al, Heart Valve Collaboratory. Standardized definitions for bioprosthetic valve dysfunction following aortic or mitral valve replacement: JACC state-of-the-art review. *J Am Coll Cardiol* 2022;80:545-61. doi:10.1016/j.jacc.2022.06.002
 - 143 Thyregod GH, Jørgensen TH, Ihlemann N, et al. Transcatheter or surgical aortic valve implantation: 10-year outcomes of the NOTION trial. *Eur Heart J* 2024;45:1116-24. doi:10.1093/eurheartj/ehae043.
 - 144 Makkak RR, Kapadia S, Chakravarty T, et al. Outcomes of repeat transcatheter aortic valve replacement with balloon-expandable valves: a registry study. *Lancet* 2023;402:1529-40. doi:10.1016/S0140-6736(23)01636-7
 - 145 Ternacle J, Pibarot P, Herrmann HC, et al. Prosthesis-patient mismatch after aortic valve replacement in the PARTNER 2 trial and registry. *JACC Cardiovasc Interv* 2021;14:1466-77. doi:10.1016/j.jcin.2021.03.069.
 - 146 Sá MP, Jacquemyn X, Van den Eynde J, et al. Impact of prosthesis-patient mismatch after transcatheter aortic valve replacement: meta-analysis of Kaplan-Meier-derived individual patient data. *JACC Cardiovasc Imaging* 2023;16:298-310. doi:10.1016/j.jcmg.2022.07.013.
 - 147 Tang GHL, Zaid S, Kleiman NS, et al. Explant vs redo-TAVR after transcatheter valve failure: mid-term outcomes from the EXPLANTORREDO-TAVR international registry. *JACC Cardiovasc Interv* 2023;16:927-41. doi:10.1016/j.jcin.2023.01.376
 - 148 Bapat VN, Zaid S, Fukuhara S, et al, EXPLANT-TAVR Investigators. Surgical explantation after TAVR failure: mid-term outcomes from the EXPLANT-TAVR international registry. *JACC Cardiovasc Interv* 2021;14:1978-91. doi:10.1016/j.jcin.2021.07.015
 - 149 Hawkins RB, Deeb GM, Sukul D, et al. Redo surgical aortic valve replacement after prior transcatheter versus surgical aortic valve replacement. *JACC Cardiovasc Interv* 2023;16:942-53. doi:10.1016/j.jcin.2023.03.015.
 - 150 Tran JH, Itagaki S, Zeng Q, et al. Transcatheter or surgical replacement for failed bioprosthetic aortic valves. *JAMA Cardiol* 2024;9:631-9. doi:10.1001/jamacardio.2024.1049
 - 151 Strom JB, Playford D, Stewart S, Strange G. An artificial intelligence algorithm for detection of severe aortic stenosis: a clinical cohort study. *JACC Adv* 2024;3:101176. doi:10.1016/j.jacadv.2024.101176.
 - 152 Cohen-Shelly M, Attia ZI, Friedman PA, et al. Electrocardiogram screening for aortic valve stenosis using artificial intelligence. *Eur Heart J* 2021;42:2885-96. doi:10.1093/eurheartj/ehab153
 - 153 Ulloa-Cerna AE, Jing L, Pfeifer JM, et al. reCHOMmed: an ECG-based machine learning approach for identifying patients at increased risk of undiagnosed structural heart disease detectable by echocardiography. *Circulation* 2022;146:36-47. doi:10.1161/CIRCULATIONAHA.121.057869
 - 154 Holste G, Oikonomou EK, Mortazavi BJ, et al. Severe aortic stenosis detection by deep learning applied to echocardiography. *Eur Heart J* 2023;44:4592-604. doi:10.1093/eurheartj/ehad456
 - 155 Wessler BS, Huang Z, Long GM Jr, et al. Automated detection of aortic stenosis using machine learning. *J Am Soc Echocardiogr* 2023;36:411-20. doi:10.1016/j.echo.2023.01.006
 - 156 Ouyang D, He B, Ghorbani A, et al. Video-based AI for beat-to-beat assessment of cardiac function. *Nature* 2020;580:252-6. doi:10.1038/s41586-020-2145-8

- 157 He B, Kwan AC, Cho JH, et al. Blinded, randomized trial of sonographer versus AI cardiac function assessment. *Nature* 2023;616:520-4. doi:10.1038/s41586-023-05947-3
- 158 Christensen M, Vukadinovic M, Yuan N, Ouyang D. Vision-language foundation model for echocardiogram interpretation. *Nat Med* 2024;30:1481-8. doi:10.1038/s41591-024-02959-y
- 159 Ueda D, Yamamoto A, Ehara S, et al. Artificial intelligence-based detection of aortic stenosis from chest radiographs. *Eur Heart J Digit Health* 2022;3:20-8. doi:10.1093/ehjdh/ztab102.
- 160 Sengupta PP, Shrestha S, Kagiya N, et al. Artificial Intelligence for Aortic Stenosis at Risk International Consortium. A machine-learning framework to identify distinct phenotypes of aortic stenosis severity. *JACC Cardiovasc Imaging* 2021;14:1707-20. doi:10.1016/j.jcmg.2021.03.020.
- 161 Krishna H, Desai K, Slostad B, et al. Fully automated artificial intelligence assessment of aortic stenosis by echocardiography. *J Am Soc Echocardiogr* 2023;36:769-77. doi:10.1016/j.echo.2023.03.008.
- 162 Strange G, Stewart S, Watts A, Playford D. Enhanced detection of severe aortic stenosis via artificial intelligence: a clinical cohort study. *Open Heart* 2023;10:e002265. doi:10.1136/openhrt-2023-002265.
- 163 Vukadinovic M, Chiu I-M, Tang X, et al. Comprehensive echocardiogram evaluation with view primed vision language AI. *Nature* 2026;650:970-7. doi:10.1038/s41586-025-09850-x
- 164 Poterucha TJ, Jing L, Ricart RP, et al. Detecting structural heart disease from electrocardiograms using AI. *Nature* 2025;644:221-30. doi:10.1038/s41586-025-09227-0
- 165 Holst K, Becker T, Magruder JT, et al. Beyond static planning: computational predictive modeling to avoid coronary artery occlusion in TAVR. *Ann Thorac Surg* 2025;119:145-51. doi:10.1016/j.athoracsur.2024.05.041
- 166 Toggweiler S, Wyler von Ballmoos MC, Moccetti F, et al. A fully automated artificial intelligence-driven software for planning of transcatheter aortic valve replacement. *Cardiovascular Revascularization Medicine* 2024;65:25-31. doi:10.1016/j.carrev.2024.03.008
- 167 Tanguturi VK, Abou-Karam R, Cheng F, et al. Electronic provider notification to facilitate the recognition and management of severe aortic stenosis: a randomized clinical trial. *Circulation* 2025;151:1498-507. doi:10.1161/CIRCULATIONAHA.125.074470.
- 168 Makkar RR, Thourani VH, Vahl TP, et al. ALIGN-AR Investigators. Transcatheter aortic valve implantation with the Trilogi valve for symptomatic native aortic regurgitation (ALIGN-AR): a pivotal, multicentre, single-arm, investigational device exemption study. *Lancet* 2025;406:2757-71. doi:10.1016/S0140-6736(25)02215-9.
- 169 Vahl TP, Thourani VH, Makkar RR, et al. Transcatheter aortic valve implantation in patients with high-risk symptomatic native aortic regurgitation (ALIGN-AR): a prospective, multicentre, single-arm study. *Lancet* 2024;403:1451-9. doi:10.1016/S0140-6736(23)02806-4.
- 170 Pawar S, Allen C, Mori M, et al. Contemporary diagnosis and treatment of aortic regurgitation: a state-of-the-art review. *J Am Coll Cardiol* 2026;87:385-413. doi:10.1016/j.jacc.2025.10.026.
- 171 Dvir D, Tchétché D, Leon MB, et al. Leaflet modification before transcatheter aortic valve implantation in patients at risk for coronary obstruction: the ShortCut study. *Eur Heart J* 2024;45:3031-41. doi:10.1093/eurheartj/ehae303.
- 172 Kodali SK, Sorajja P, Meduri CU, et al. Early safety and feasibility of a first-in-class biomimetic transcatheter aortic valve - DurAVR. *EuroIntervention* 2023;19:e352-62. doi:10.4244/EIJ-D-23-00282
- 173 George I, Rao DP, Jain A, et al. 1-year results from a multicenter trial of a polymer surgical mitral valve: insights into new technology. *J Am Coll Cardiol* 2025;86:515-26. doi:10.1016/j.jacc.2025.06.017.
- 174 Stinis CT, Abbas AE, Teirstein P, et al. Real-world outcomes for the fifth-generation balloon expandable transcatheter heart valve in the United States. *JACC Cardiovasc Interv* 2024;17:1032-44. doi:10.1016/j.jcin.2024.02.015.
- 175 Fezzi S, Jagielak D, Targonski R, et al. Final report of the PROTEMBO C Trial: a prospective evaluation of a novel cerebral protection device during TAVI. *EuroIntervention* 2024;20:e264-7. doi:10.4244/EIJ-D-23-00743
- 176 Ng VG, Leon MB. The Future of Transcatheter Aortic Valve Replacement. *Transcatheter Aortic Valve Implantation*. CRC Press, 2025:355-63.
- 177 Shah BN. National Institute for Health and Care Excellence (NICE) guidance on heart valve disease. *Heart* 2023;109:817-22. doi:10.1136/heartjnl-2022-321095

Web appendix: PRISMA flow diagram